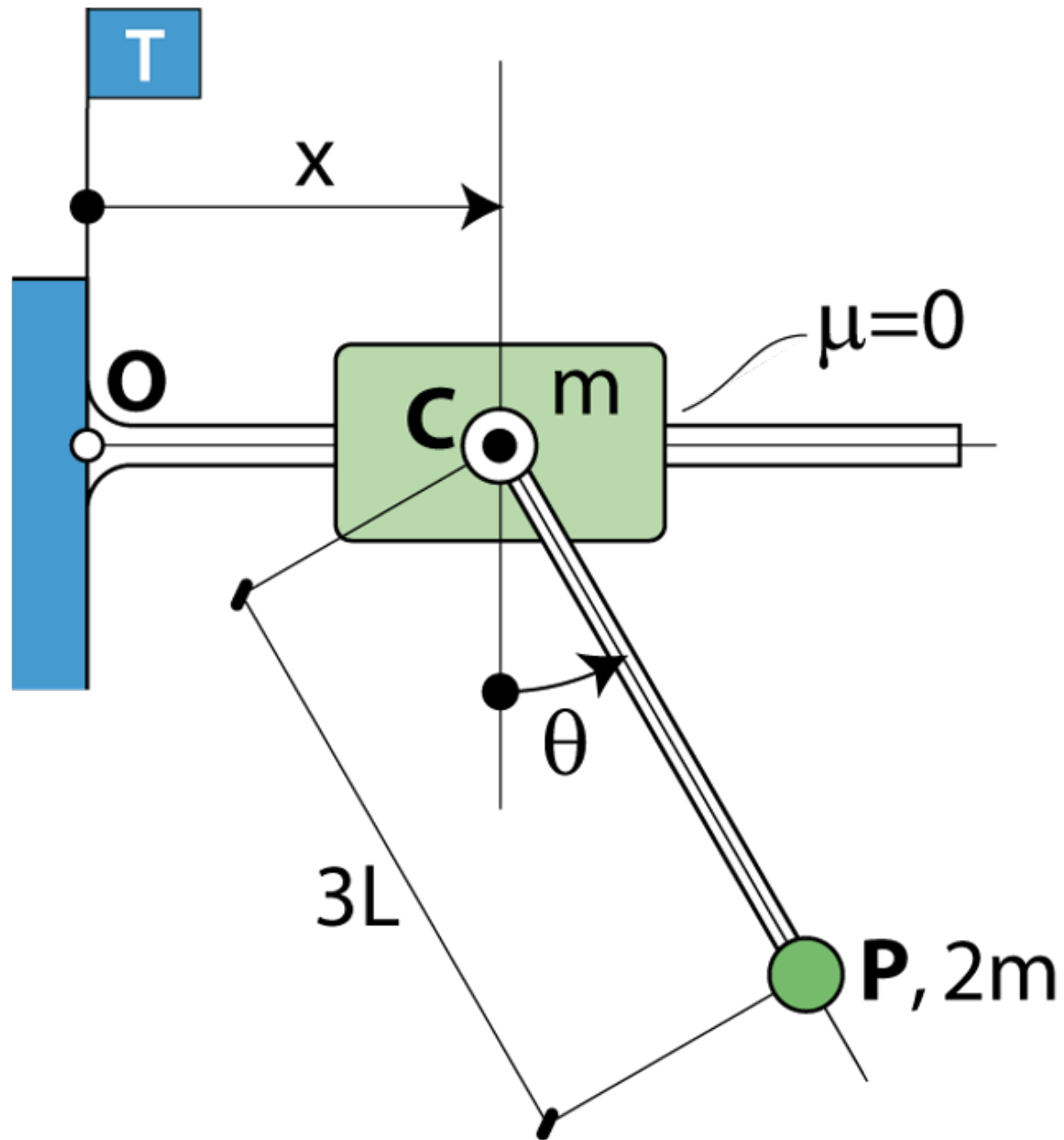


11P

Teoremes vectorials II

Exemples 3D



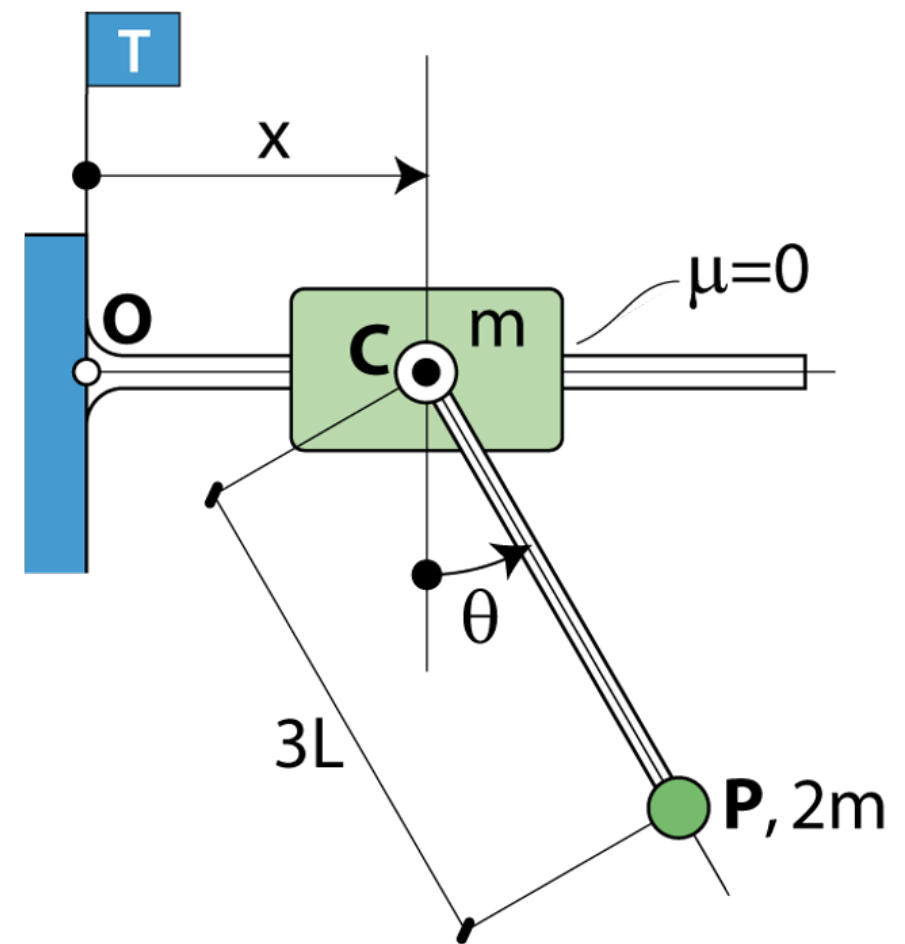
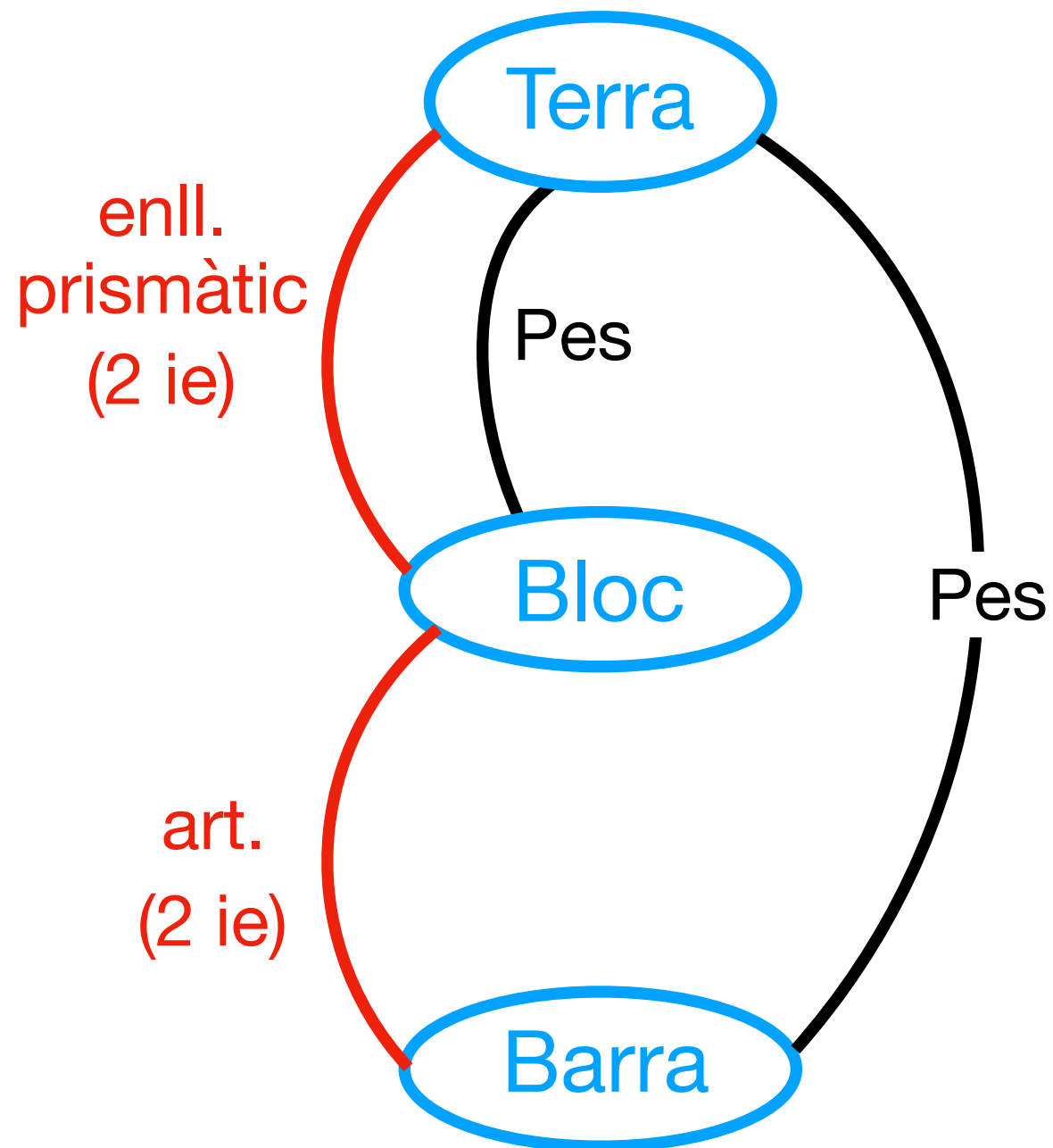
- Equacions del moviment
- $x(t)$ si es posa un motor entre bloc i pèndol que garanteix $\dot{\theta} = \omega_0 = ct$

Condicions inicials	$x(0) = 0$ $\dot{x}(0) = 0$ $\theta(0) = 0$
------------------------	---

Tractat com de
dinàmica 2D

DGI

(sense motor)

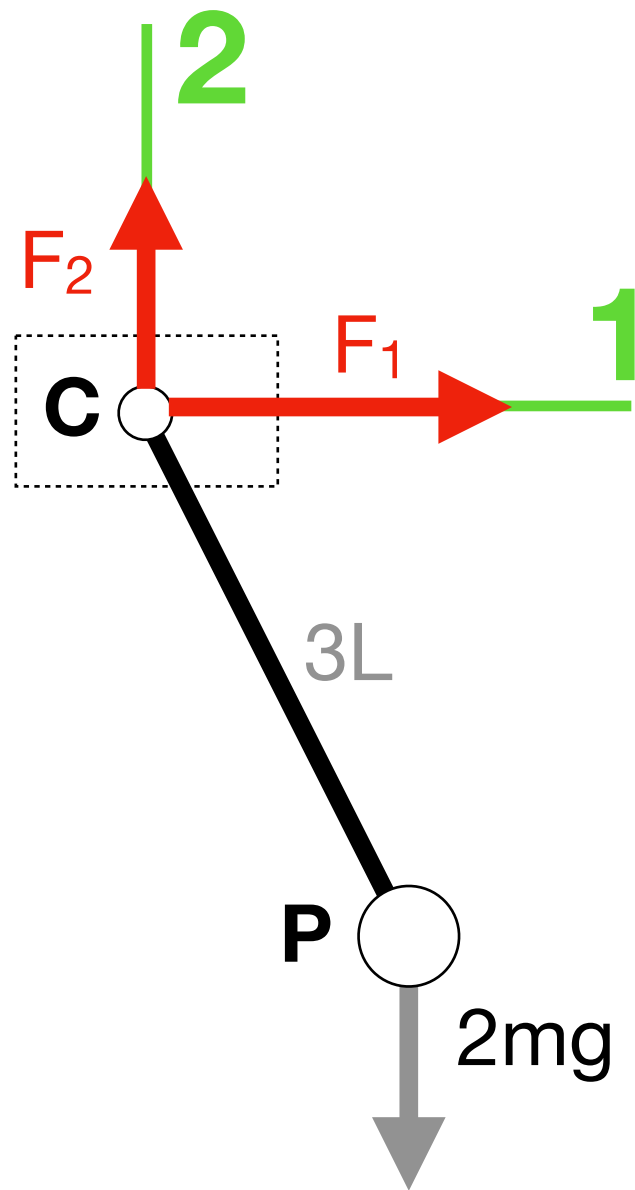


Problema dinàmic global

6 eqs = 2 sòlids · 3 eqs/sòlid

6 incògn. $\left| \begin{array}{l} 4 \text{ i.e.} \\ \ddot{x}, \ddot{\theta} \\ \text{Associades} \\ \text{a GL} \end{array} \right.$

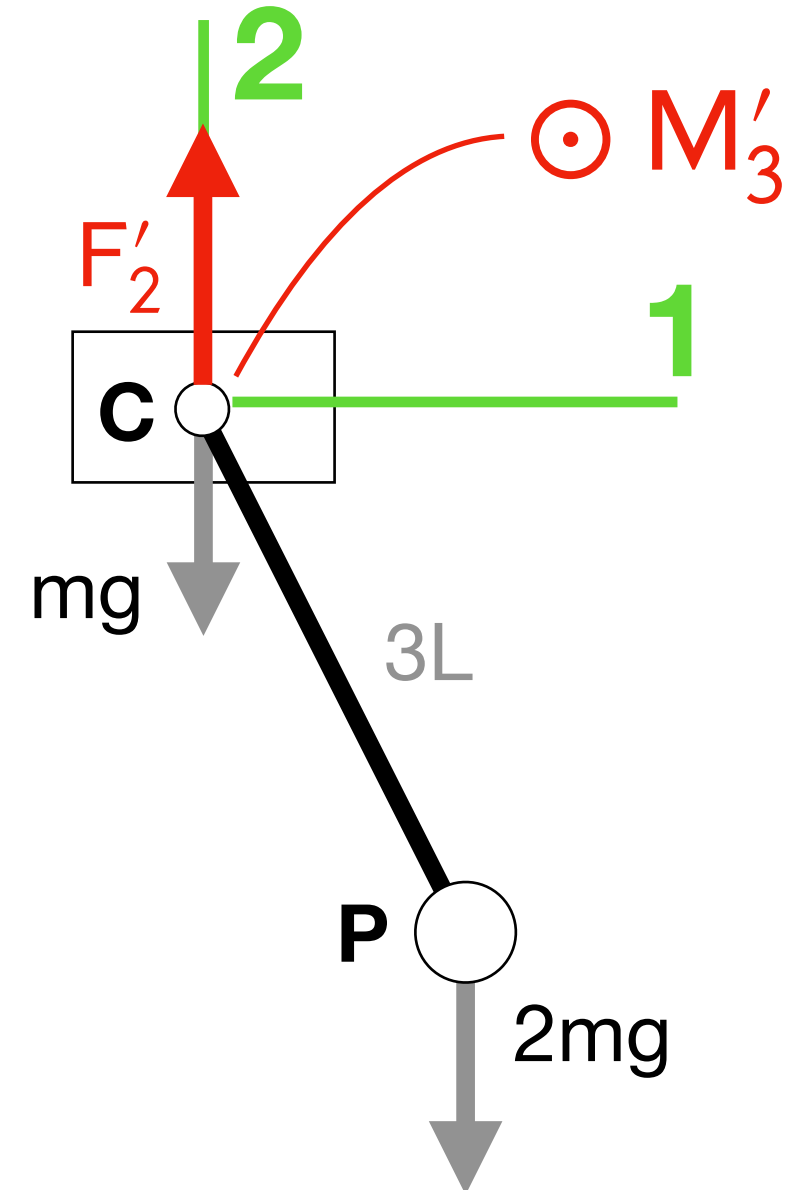
Sist = Barra



$$\left\{ \bar{\mathbf{F}}_{\text{Bloc} \rightarrow \text{Barra}} \right\}_B = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix}$$

$$\left\{ \bar{\mathbf{M}}_{\text{Bloc} \rightarrow \text{Barra}} (\mathbf{C}) \right\}_B = \{0\}$$

Sist = Bloc + Barra

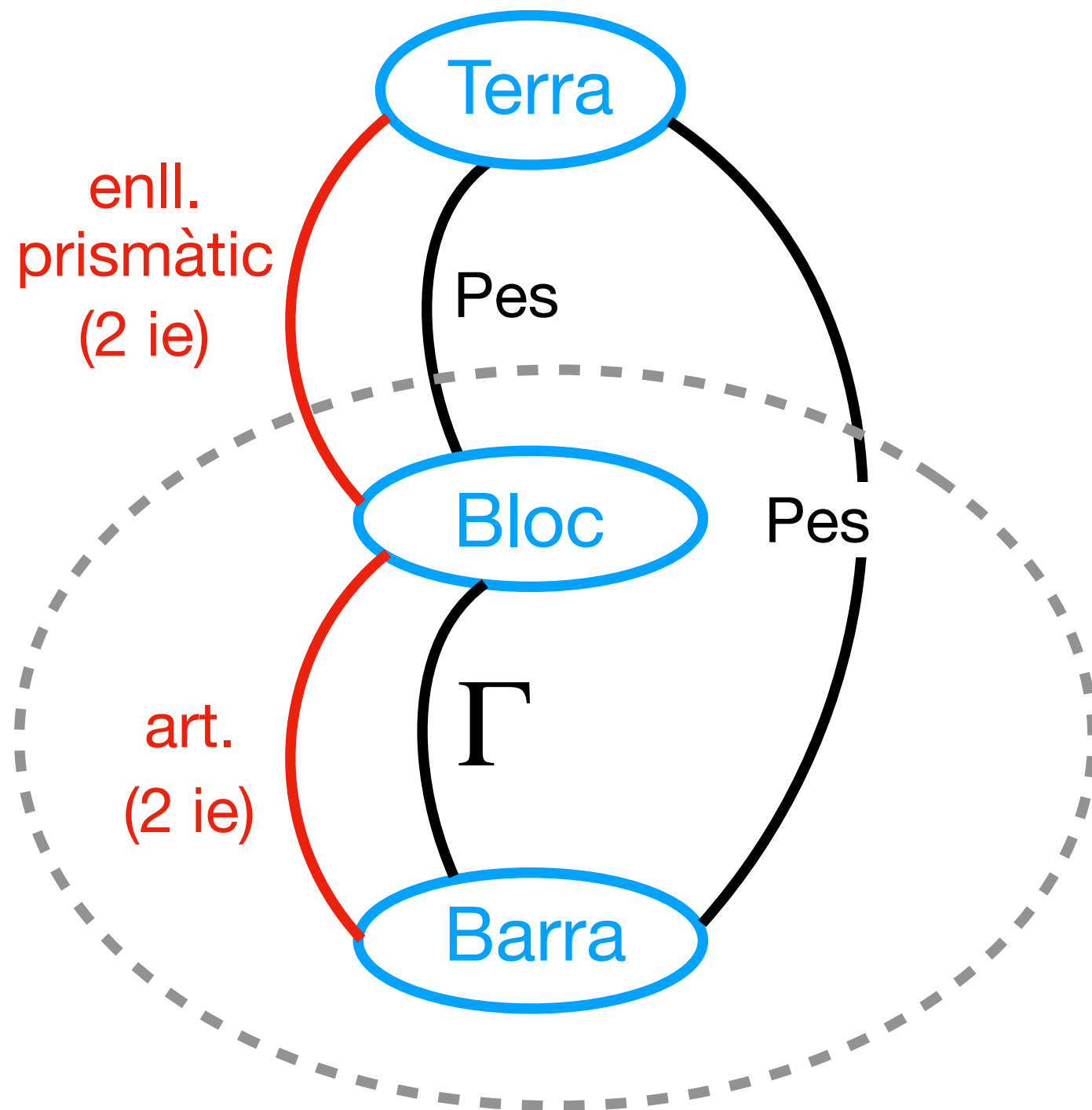


$$\left\{ \bar{\mathbf{F}}_{\text{T} \rightarrow \text{Bloc}} \right\}_B = \begin{Bmatrix} 0 \\ F'_2 \end{Bmatrix}$$

$$\left\{ \bar{\mathbf{M}}_{\text{T} \rightarrow \text{Bloc}} (\mathbf{C}) \right\}_B = \{M'_3\}$$

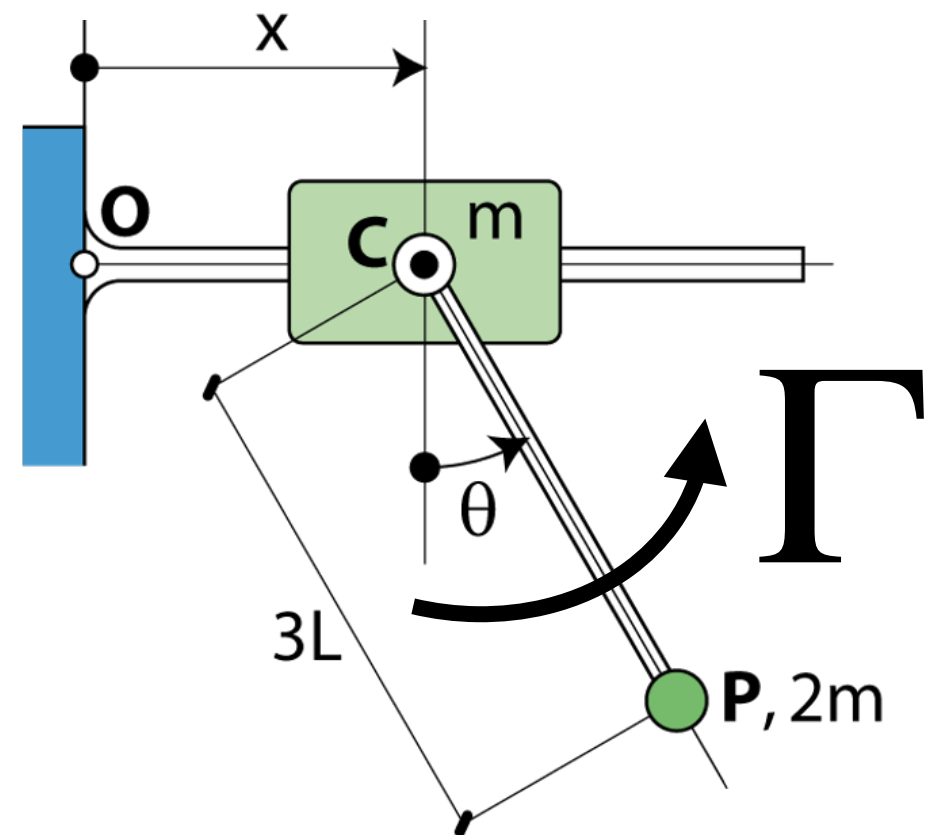
DGI

(amb motor)

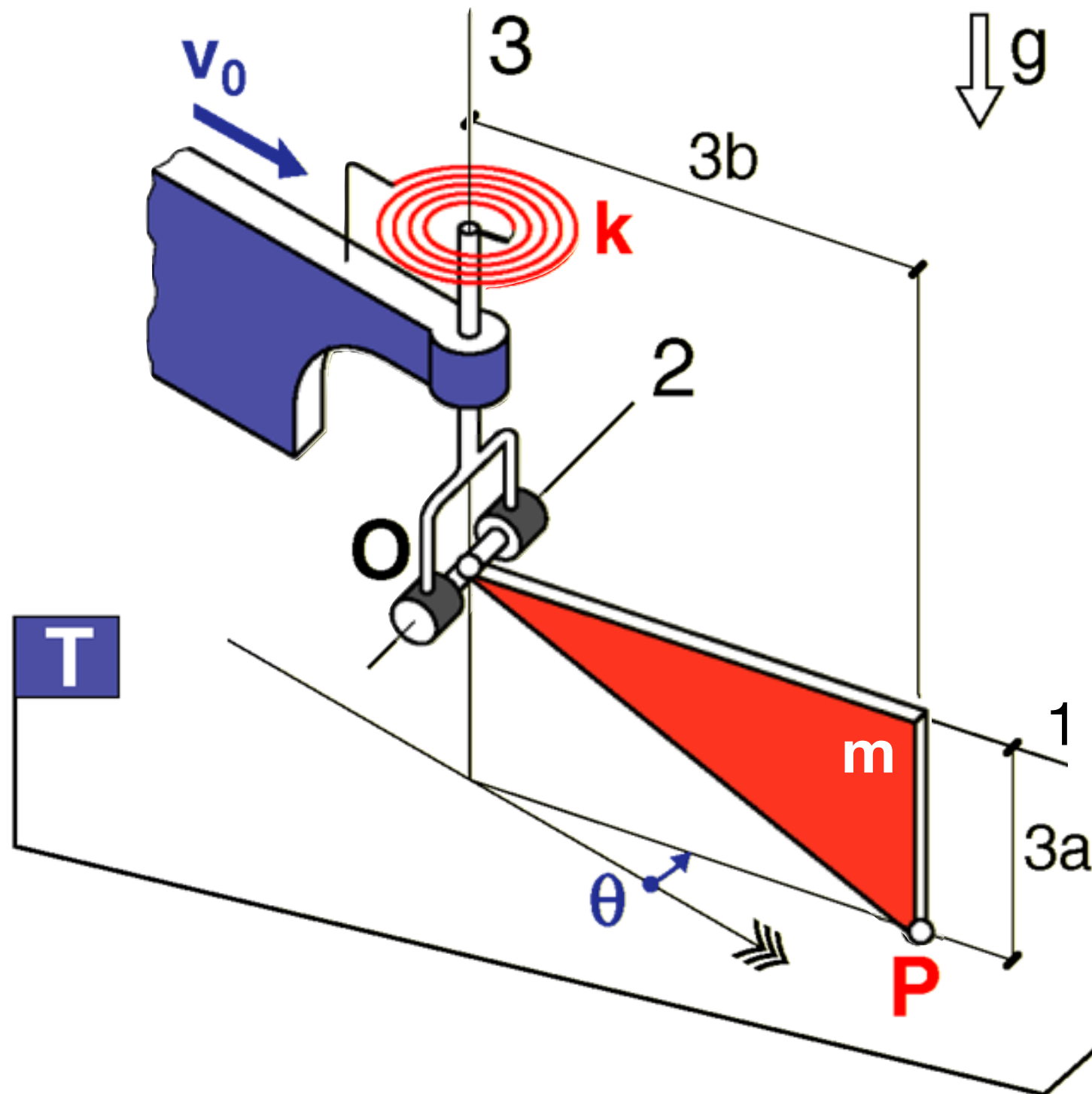


Ara

Hi ha el parell estàtor → rotor



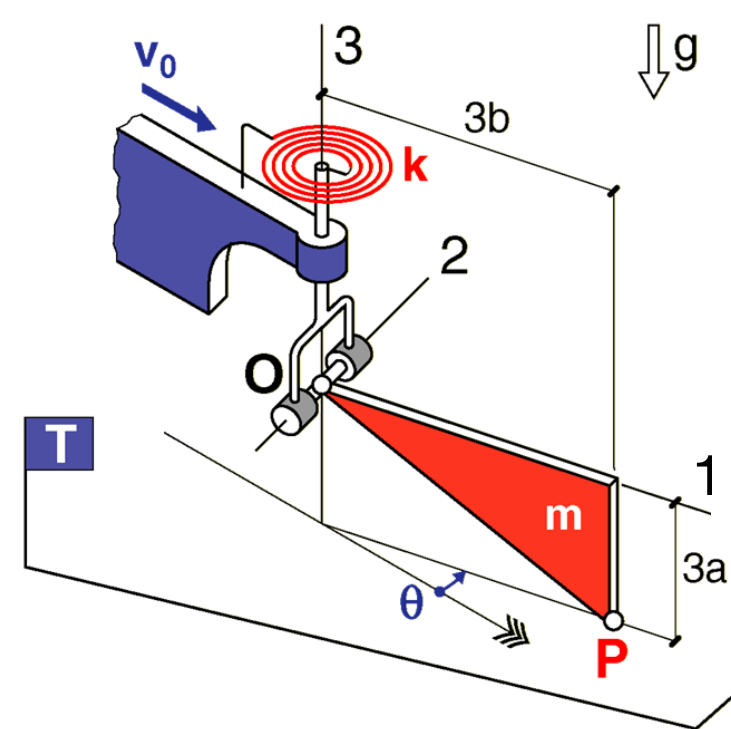
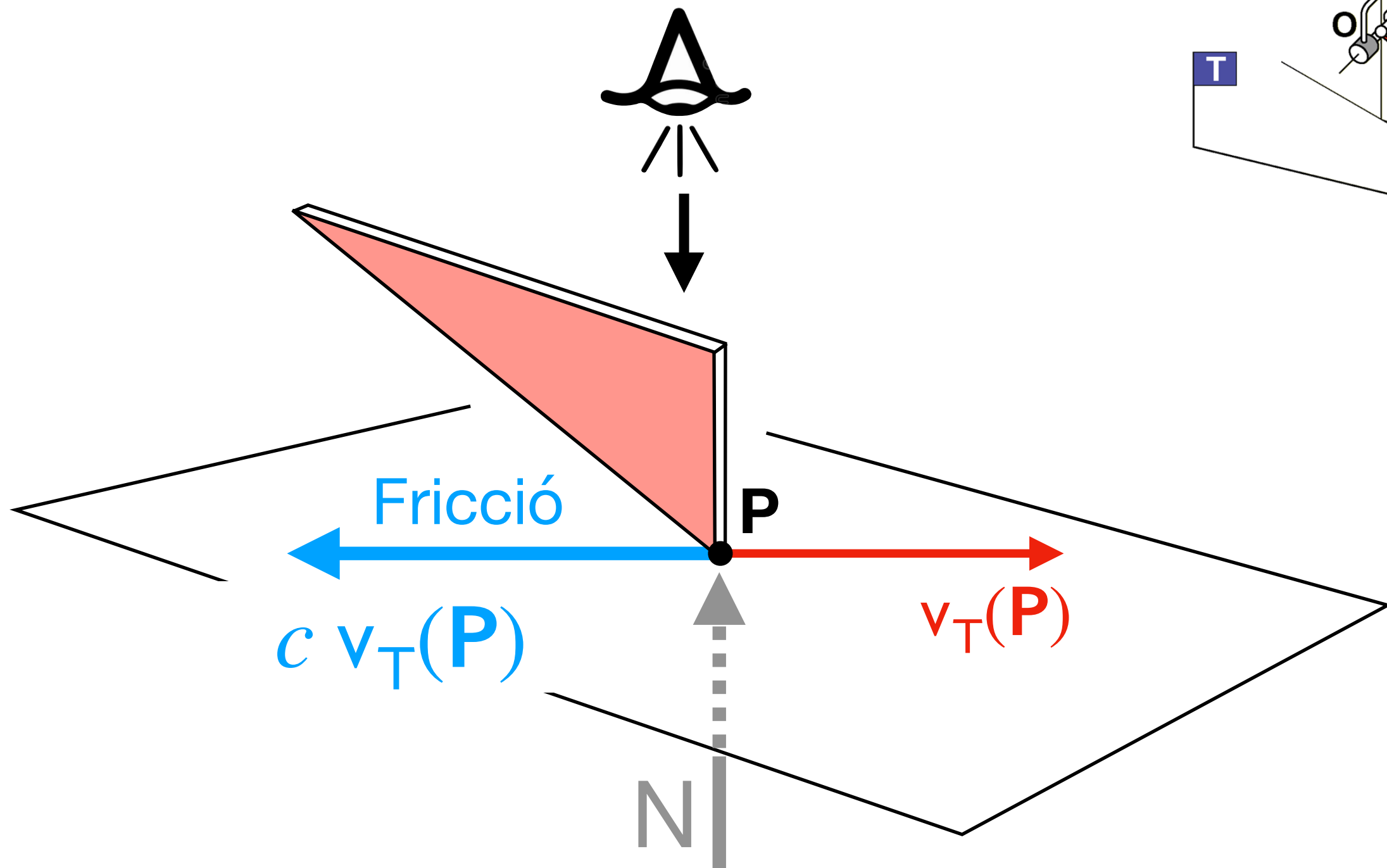
- Eq. mov. per a θ ?
- $k_{\min}$ per a que $\theta_{eq} = 0$ sigui **ESTABLE** ?



$\exists$ freq viscós $T \rightarrow \mathbf{P}$
(de coef c)

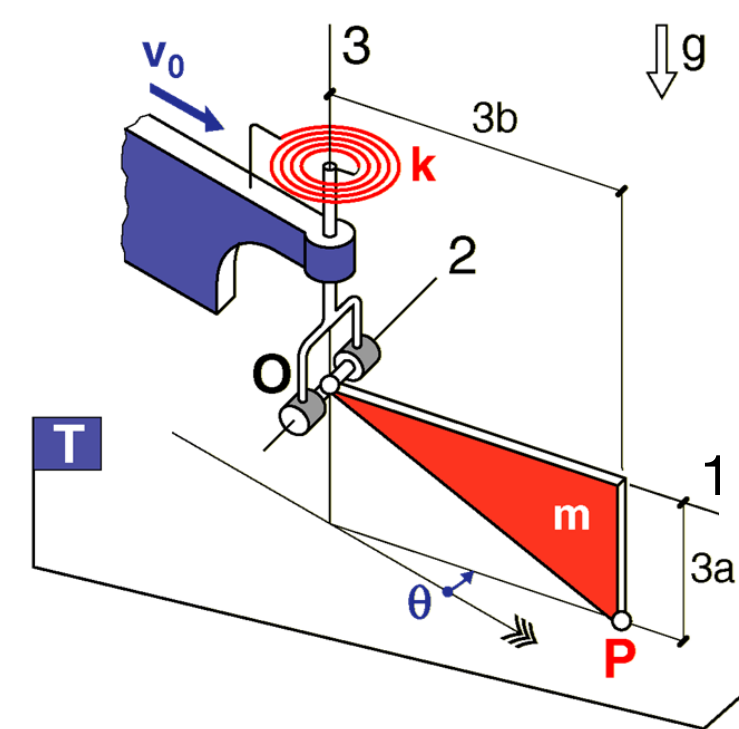
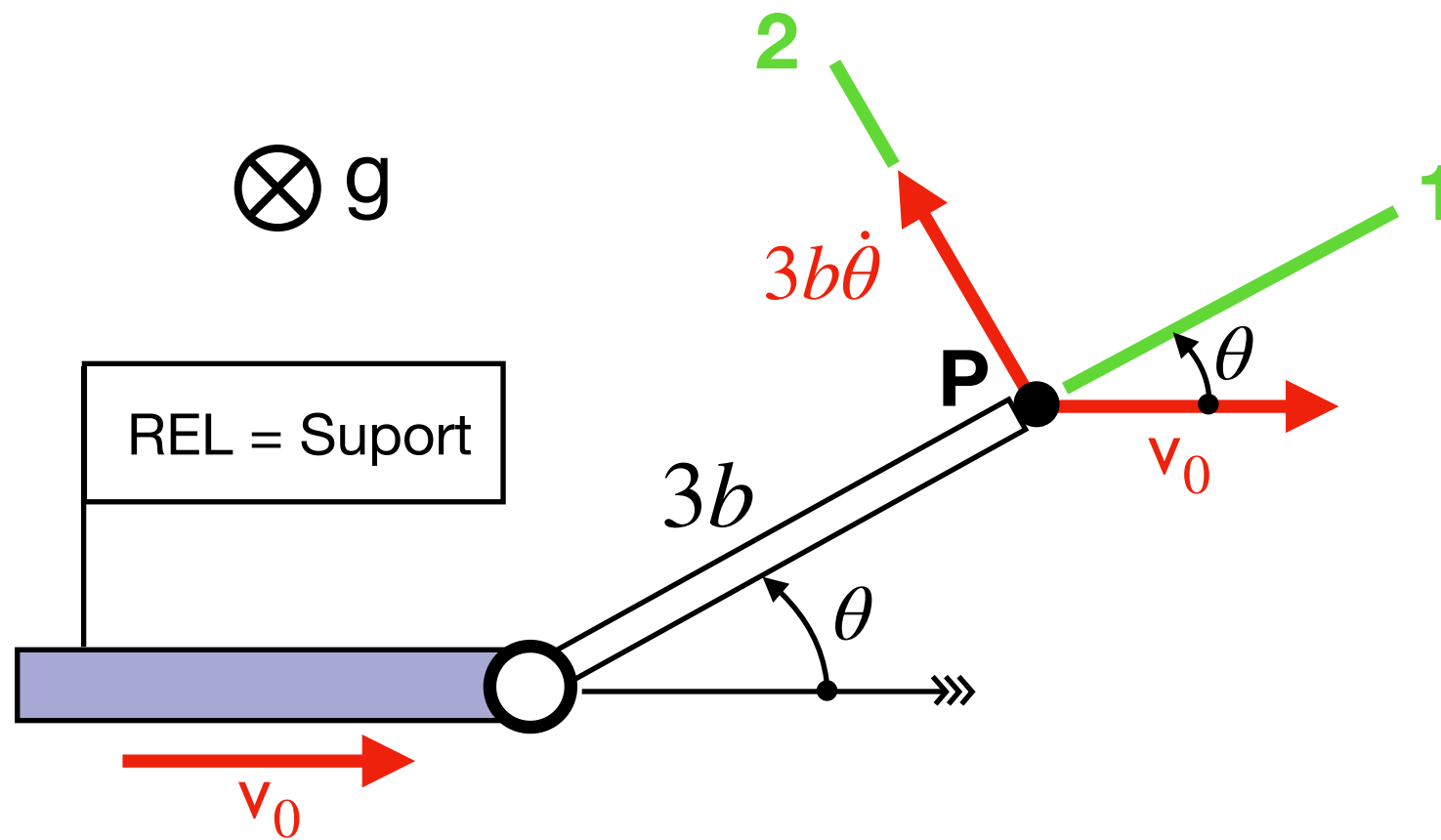
Per $\theta = 0$ la molla
està distesa

Força de frec viscós $T \rightarrow P$



$$\bar{F}_{fv} = -c \bar{v}_T(P)$$

Força de frec viscós $T \rightarrow P$

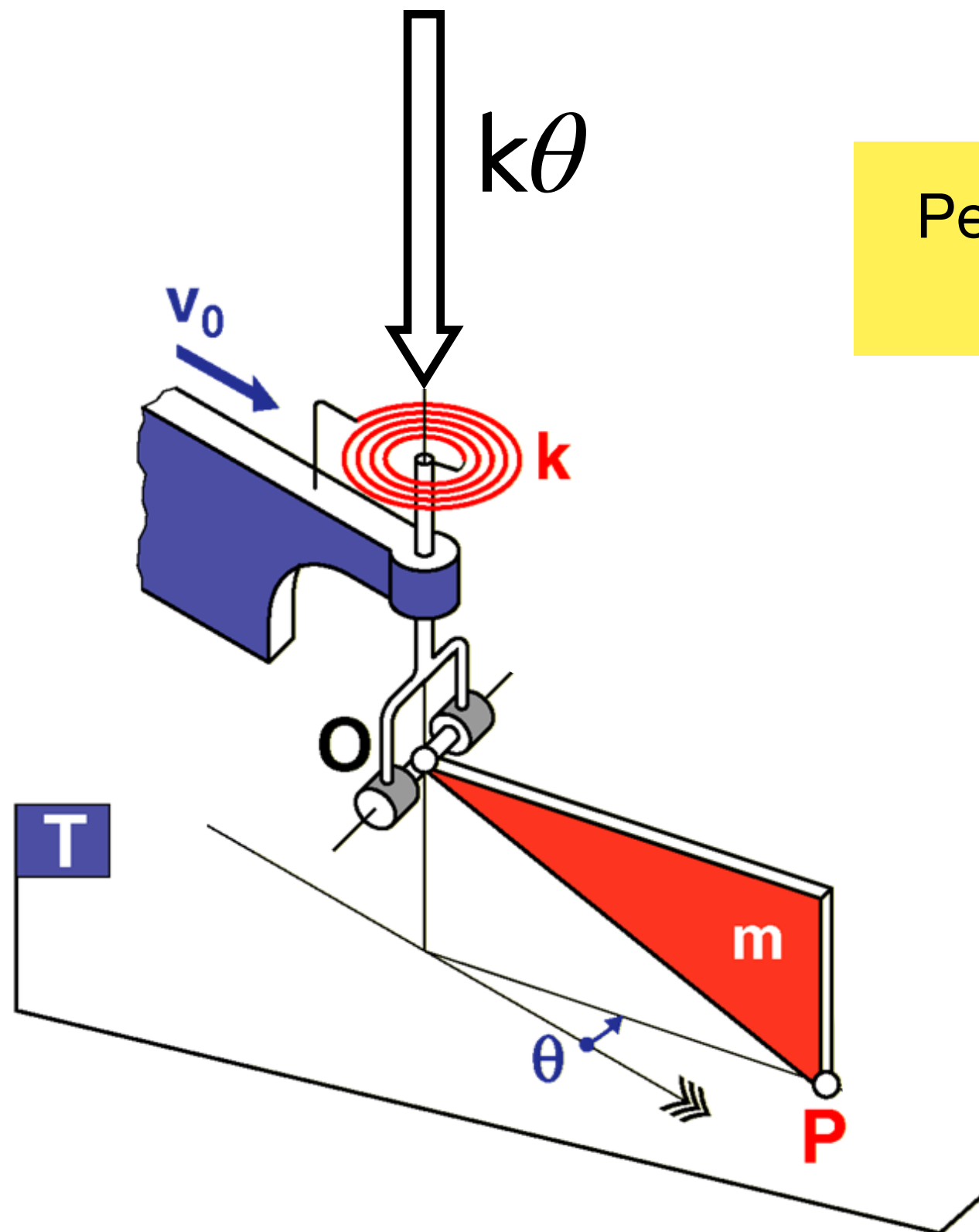


$$\bar{v}_T(\mathbf{P}) = \bar{v}_{REL}(\mathbf{P}) + \bar{v}_{ar}(\mathbf{P}) = \begin{Bmatrix} v_0 \cos \theta \\ -v_0 \sin \theta + 3b\dot{\theta} \\ 0 \end{Bmatrix}_{B=(1,2,3)}$$

$$\bar{F}_{fv} = -c \bar{v}_T(\mathbf{P}) = \begin{Bmatrix} -cv_0 \cos \theta \\ cv_0 \sin \theta - 3cb\dot{\theta} \\ 0 \end{Bmatrix}_B$$

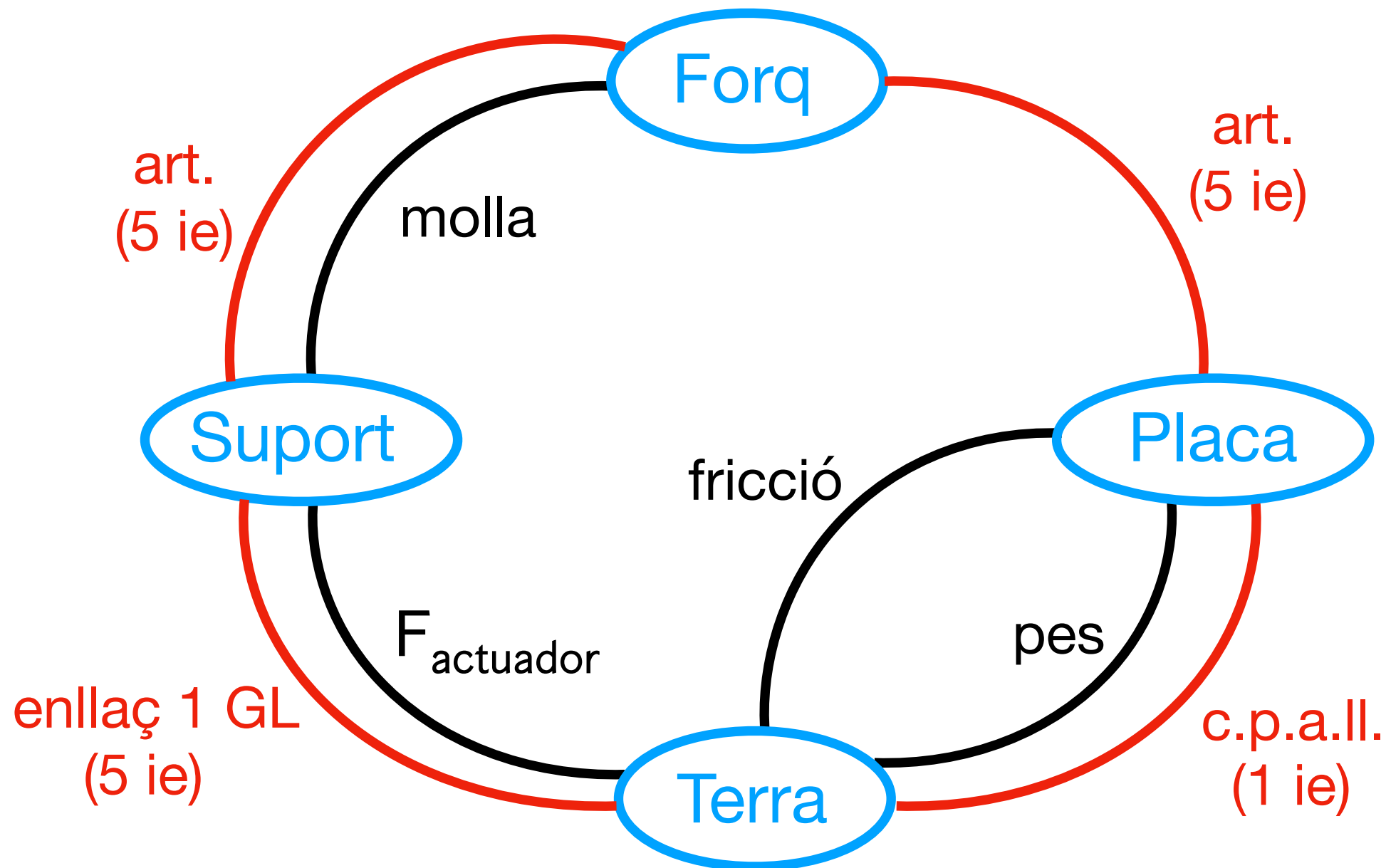
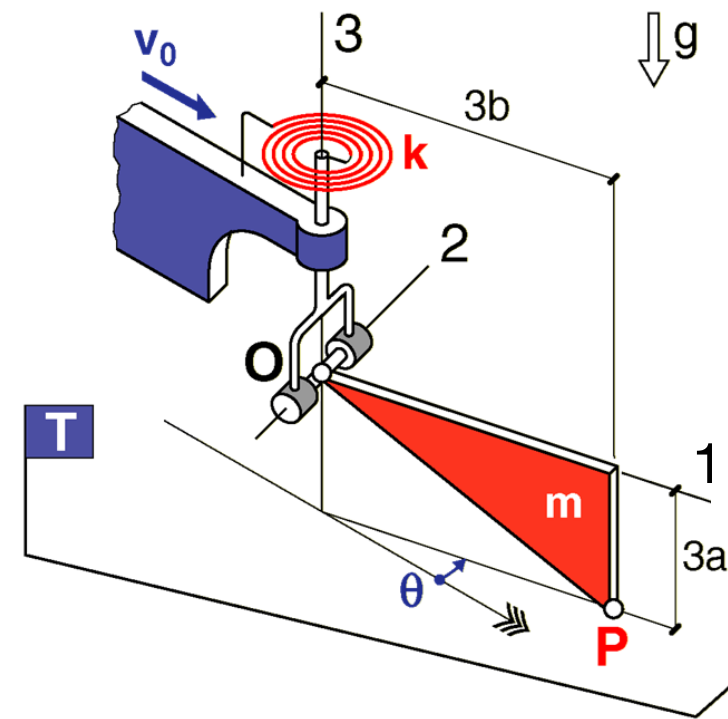
F_{fv1}
 F_{fv2}

Parell molla torsional $\rightarrow$ forq



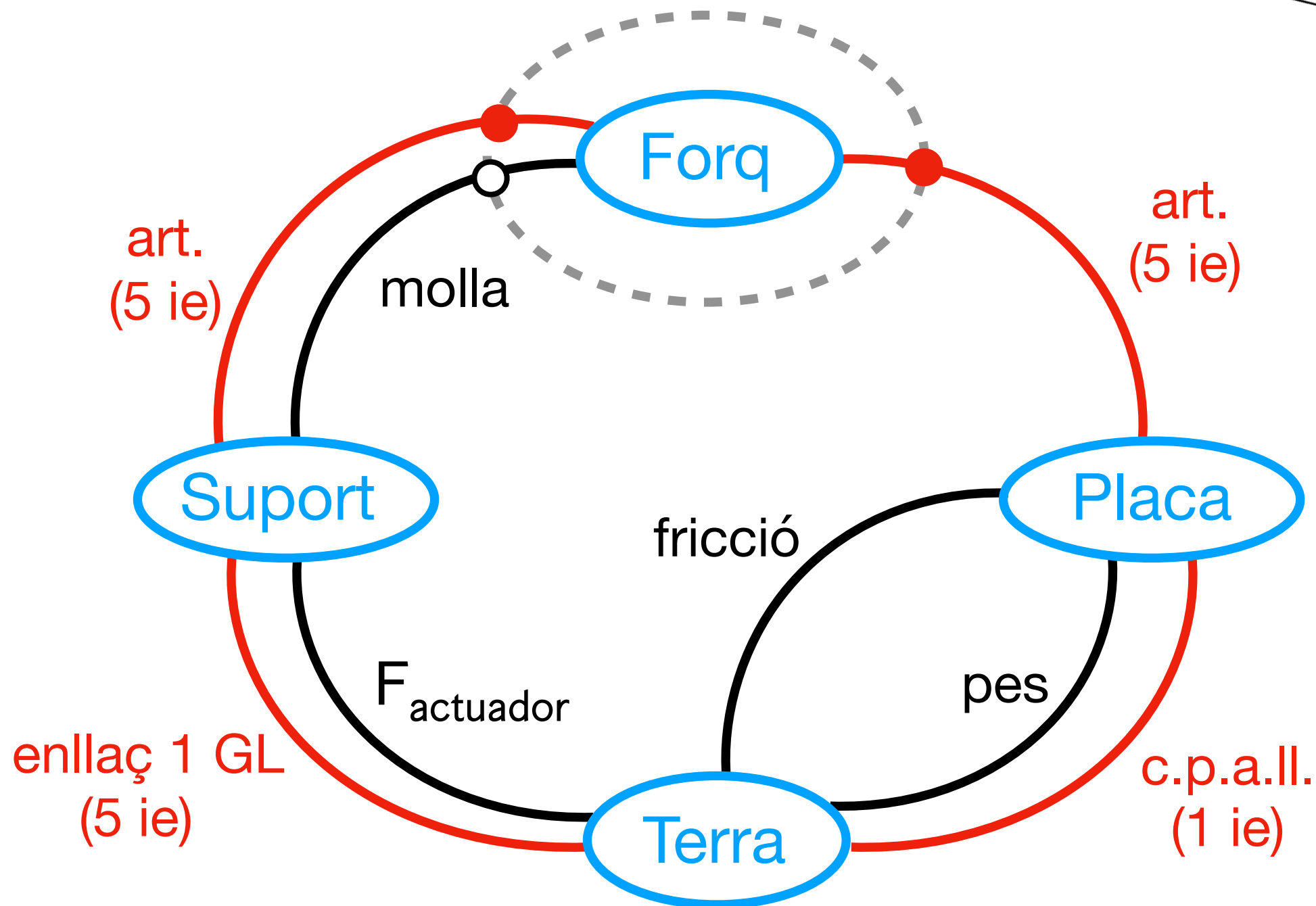
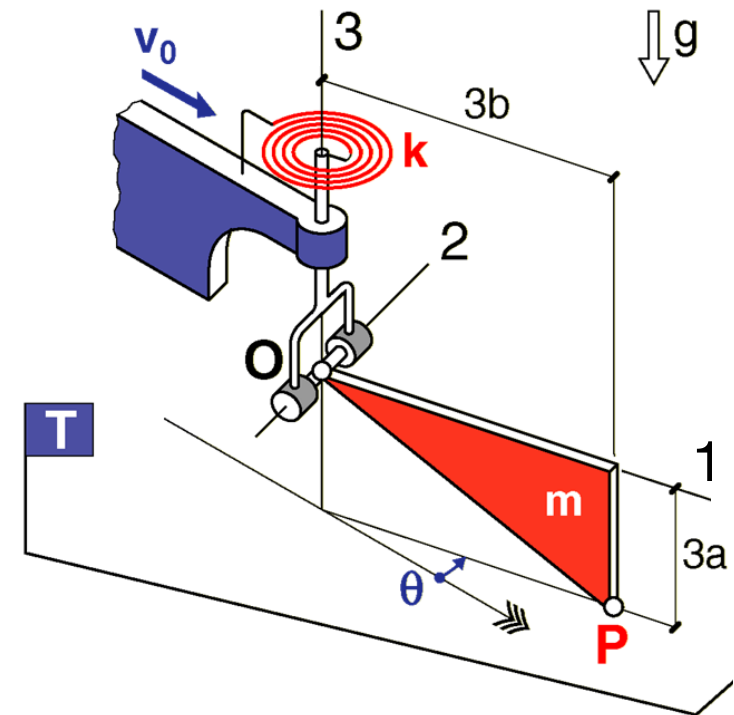
Per $\theta = 0$ la molla
està distesa

DGI = Diagrama general d'interaccions



INDET

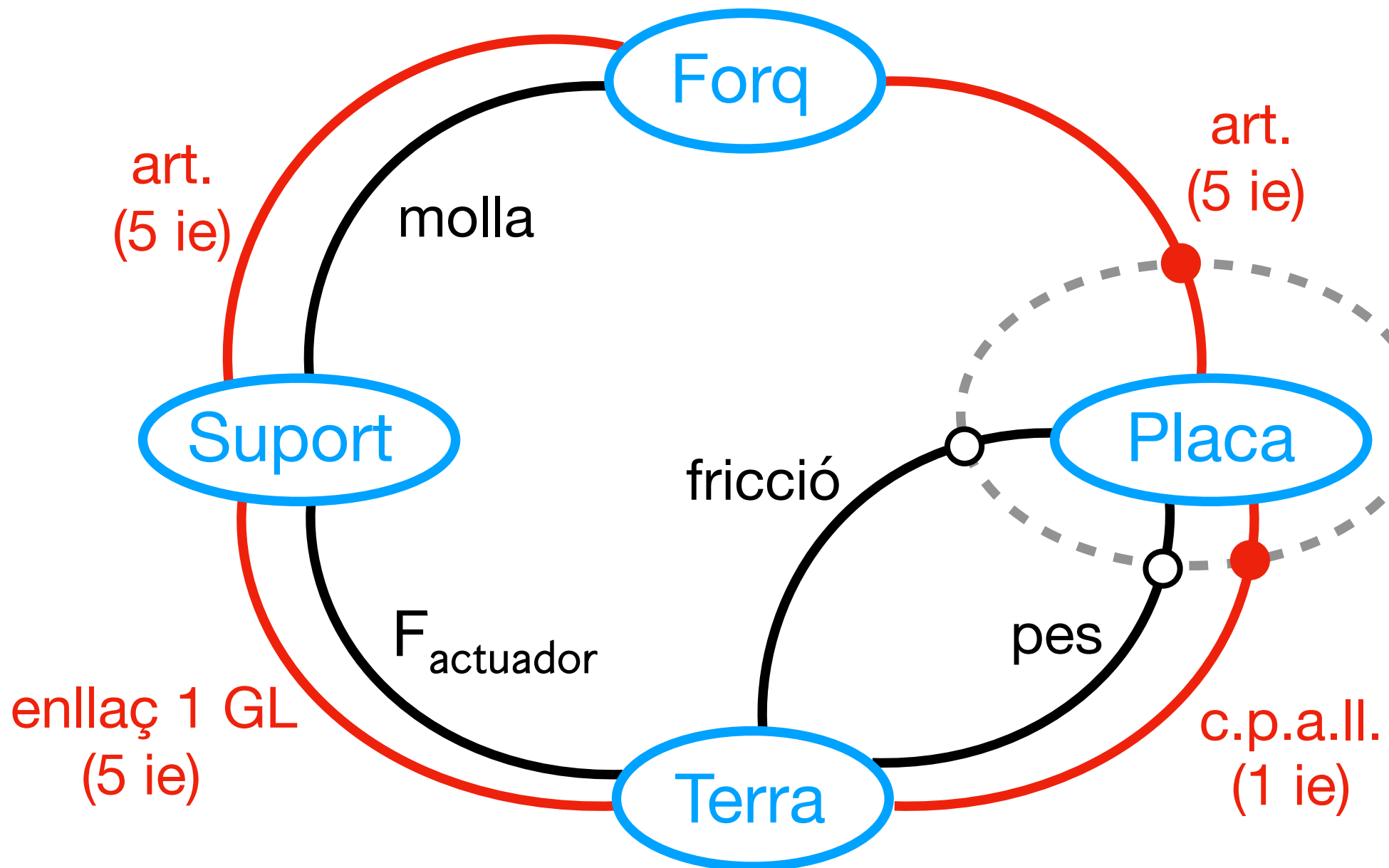
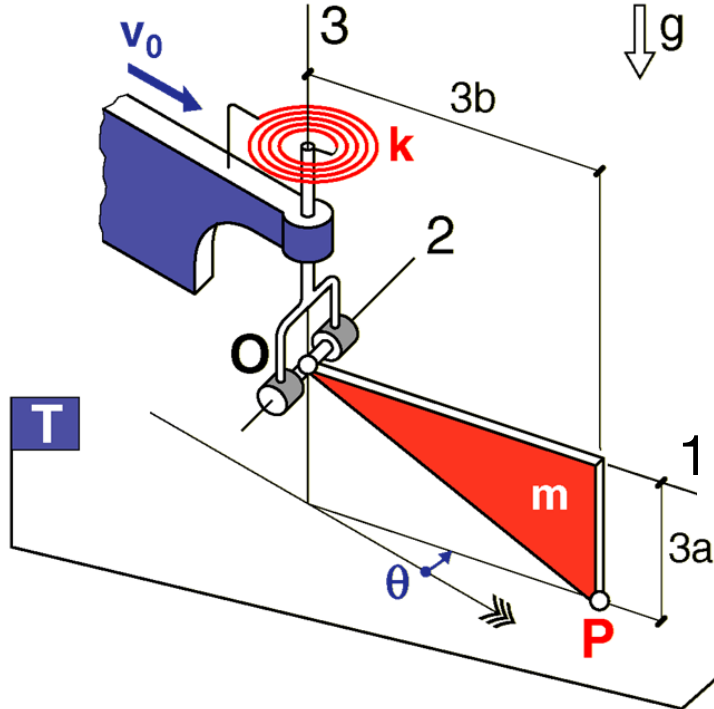
$$\left\{ \begin{array}{l} \text{Sist} = \text{Forq} \\ 10 \text{ ie} + \ddot{\theta} = 11 \text{ incòg} \end{array} \right.$$



INDET

Sist = Placa
6 ie + $\ddot{\theta}$ = 7 incòg

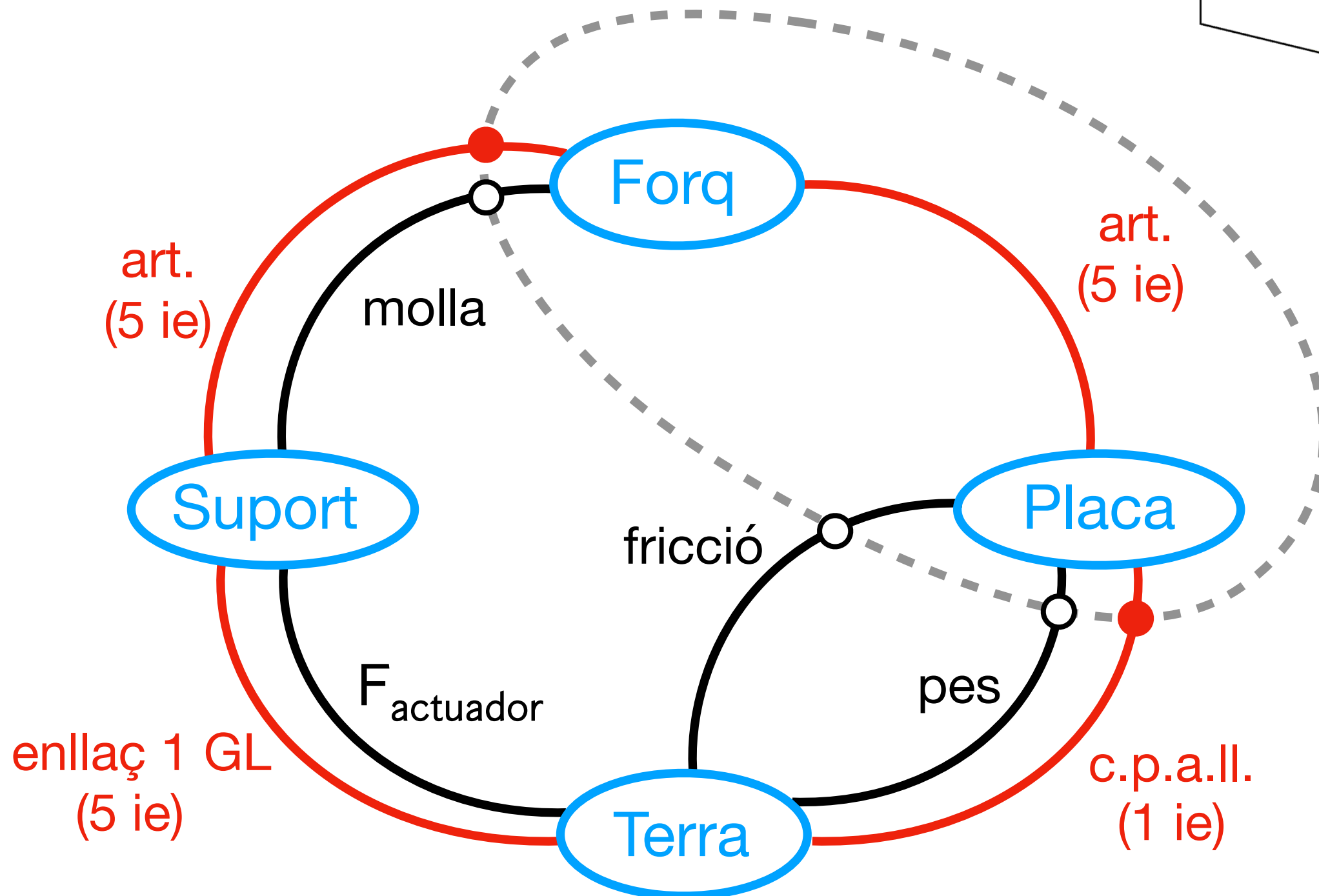
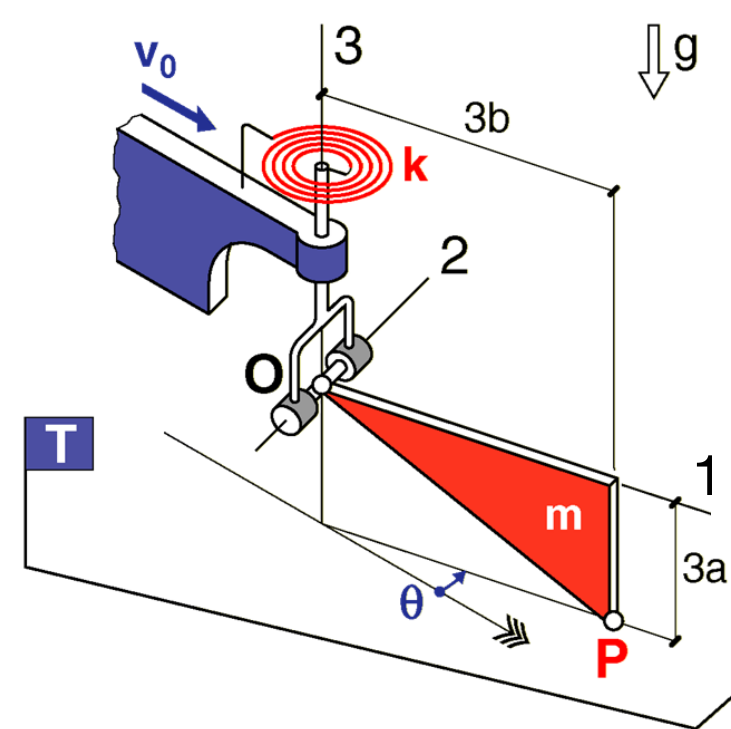
6 ie + $\ddot{\theta}$ = 7 incòg



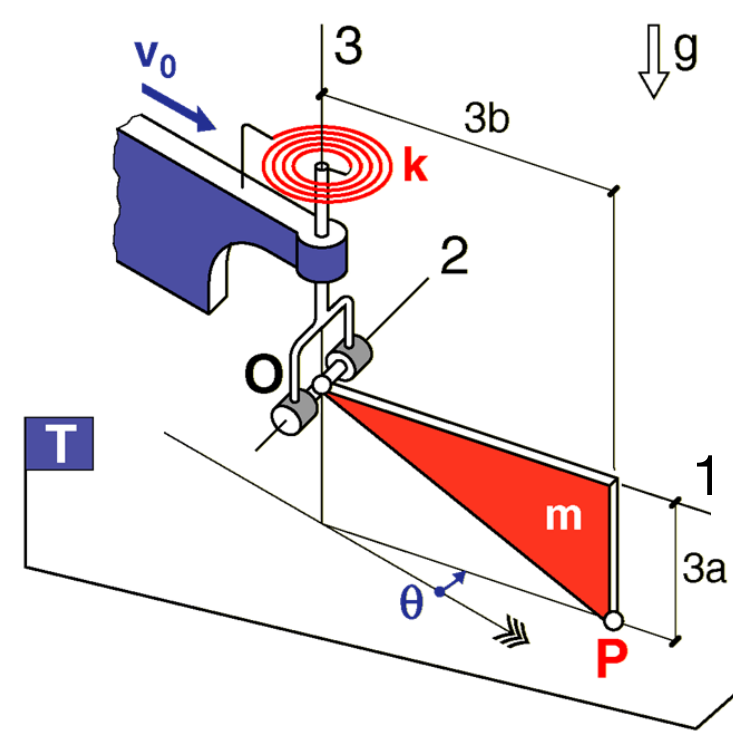
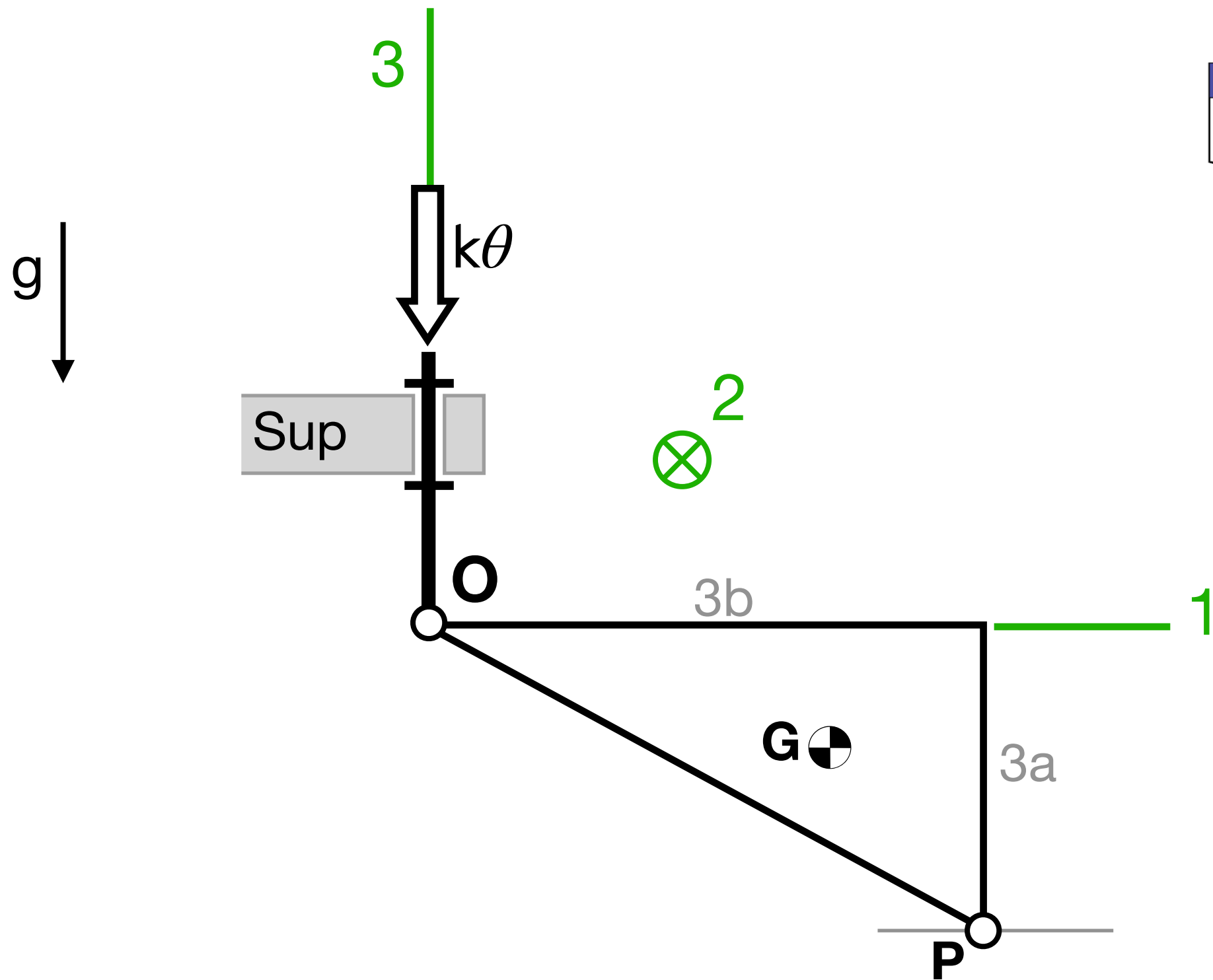
INDET

Sist = Placa + Forq

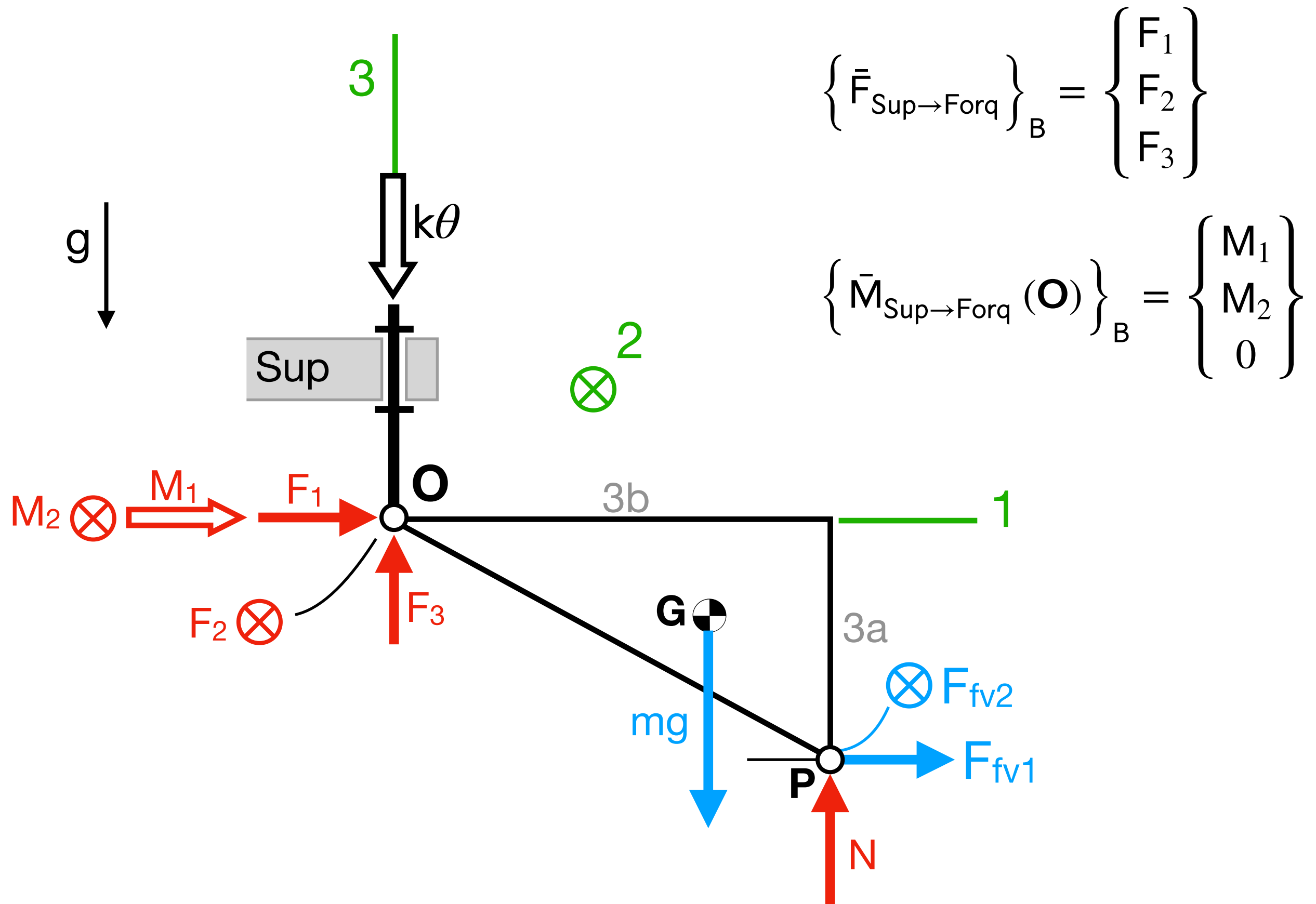
$$6 \text{ ie} + \ddot{\theta} = 7 \text{ incòg}$$



Forces sobre "Placa + Forq"



Forces sobre "Placa + Forq"



Anàlisi de l'estabilitat de $\theta_{eq} = 0$

3 passos
com al pèndol simple

$$I_{33} \ddot{\theta} + 9cb^2 \dot{\theta} + k\theta - 3bcv_0 \sin \theta = 0$$

Obtenim EDO de l'error ε

$$\theta = \theta_{eq} + \varepsilon = \varepsilon$$

$$\dot{\theta} = \dot{\varepsilon}$$

$$\ddot{\theta} = \ddot{\varepsilon}$$

en aquest exercici

$$I_{33} \ddot{\varepsilon} + 9cb^2 \dot{\varepsilon} + k\varepsilon - 3bcv_0 \sin \varepsilon = 0$$

La linealitzem

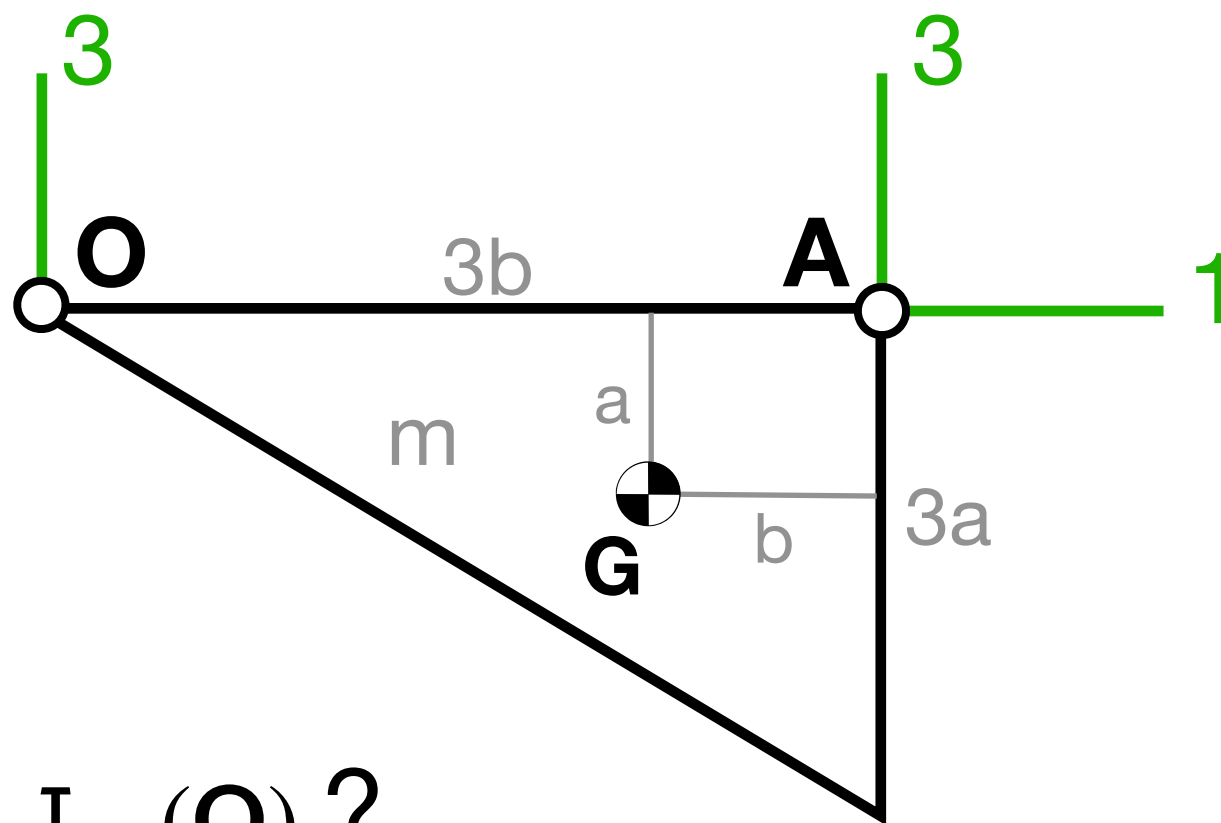
$$\sin \varepsilon \approx \varepsilon$$

$$I_{33} \ddot{\varepsilon} + \underbrace{9cb^2}_A \dot{\varepsilon} + \underbrace{(k - 3bcv_0)}_B \varepsilon = 0$$

$$\ddot{\varepsilon} = -\underbrace{\frac{B}{I_{33}}}_K \varepsilon - \underbrace{\frac{A}{I_{33}}}_{C > 0} \dot{\varepsilon}$$

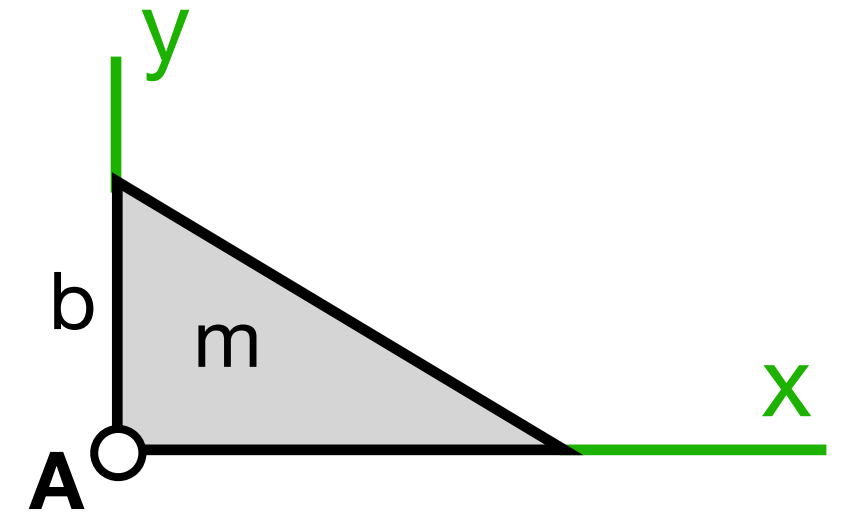
$K > 0?$

$$K > 0 \iff B > 0 \iff k > 3bcv_0$$



$I_{33}(\mathbf{O})$?

Taules



$$I_{xx}(\mathbf{A}) = \frac{1}{6}mb^2$$

$I_{33}(\mathbf{A})$ de taules + **doble Steiner** per passar de **A** a **O**:

$$(a) \quad I_{33}(\mathbf{O}) = I_{33}(\mathbf{G}) + I_{33}^{\oplus}(\mathbf{O})$$

$$(b) \quad I_{33}(\mathbf{A}) = I_{33}(\mathbf{G}) + I_{33}^{\oplus}(\mathbf{A})$$

$$(a - b) \quad I_{33}(\mathbf{O}) = I_{33}(\mathbf{A}) + I_{33}^{\oplus}(\mathbf{O}) - I_{33}^{\oplus}(\mathbf{A})$$

$$I_{33}(\mathbf{O}) = \frac{1}{6}m(3b)^2 + m(2b)^2 - mb^2 = \frac{9}{2}mb^2$$

DEURES

Determineu el valor de la normal N del terra sobre la placa en funció de les variables d'estat del sistema

Pistes

- Apliqueu el full de ruta

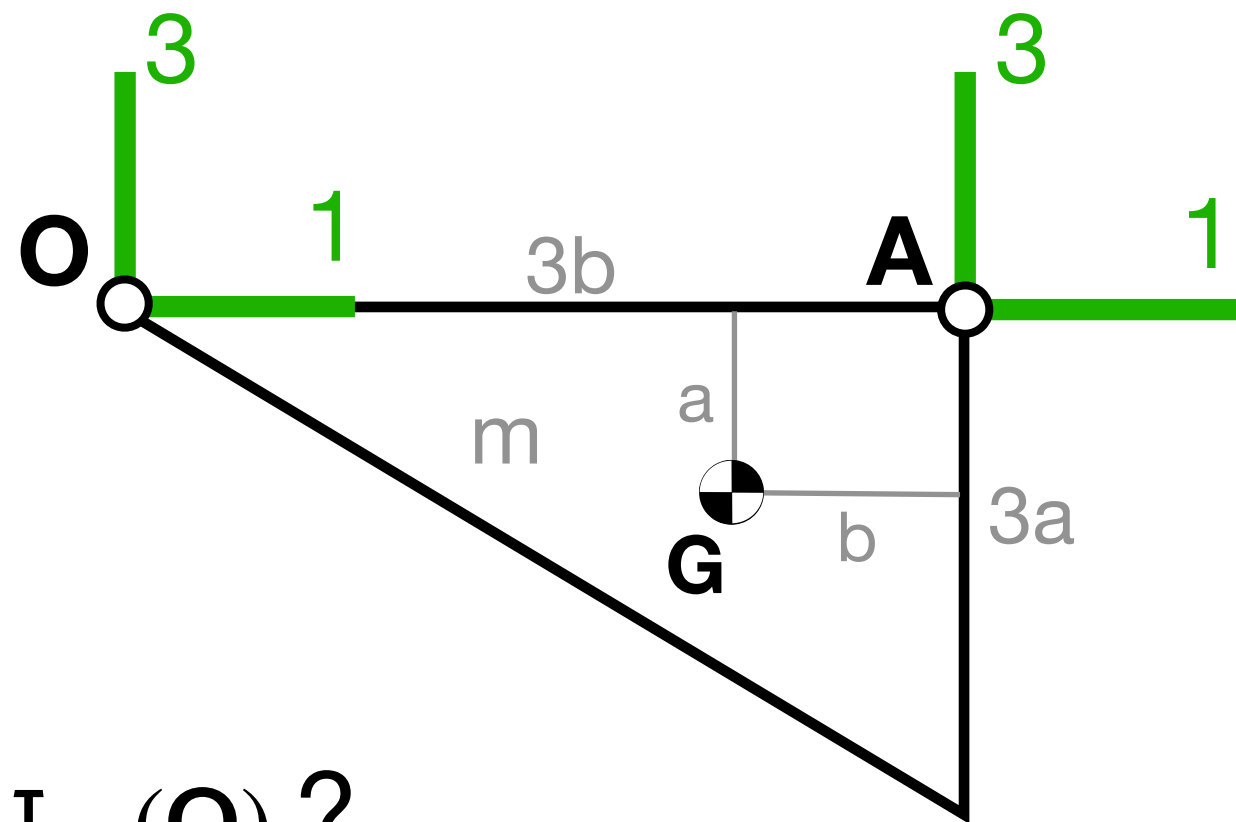
SISTEMA = Placa

$\text{TMC}(\mathbf{O})\big]_2$ (aprofiteu el moment cinètic calculat abans per a l'eq. del mov.)

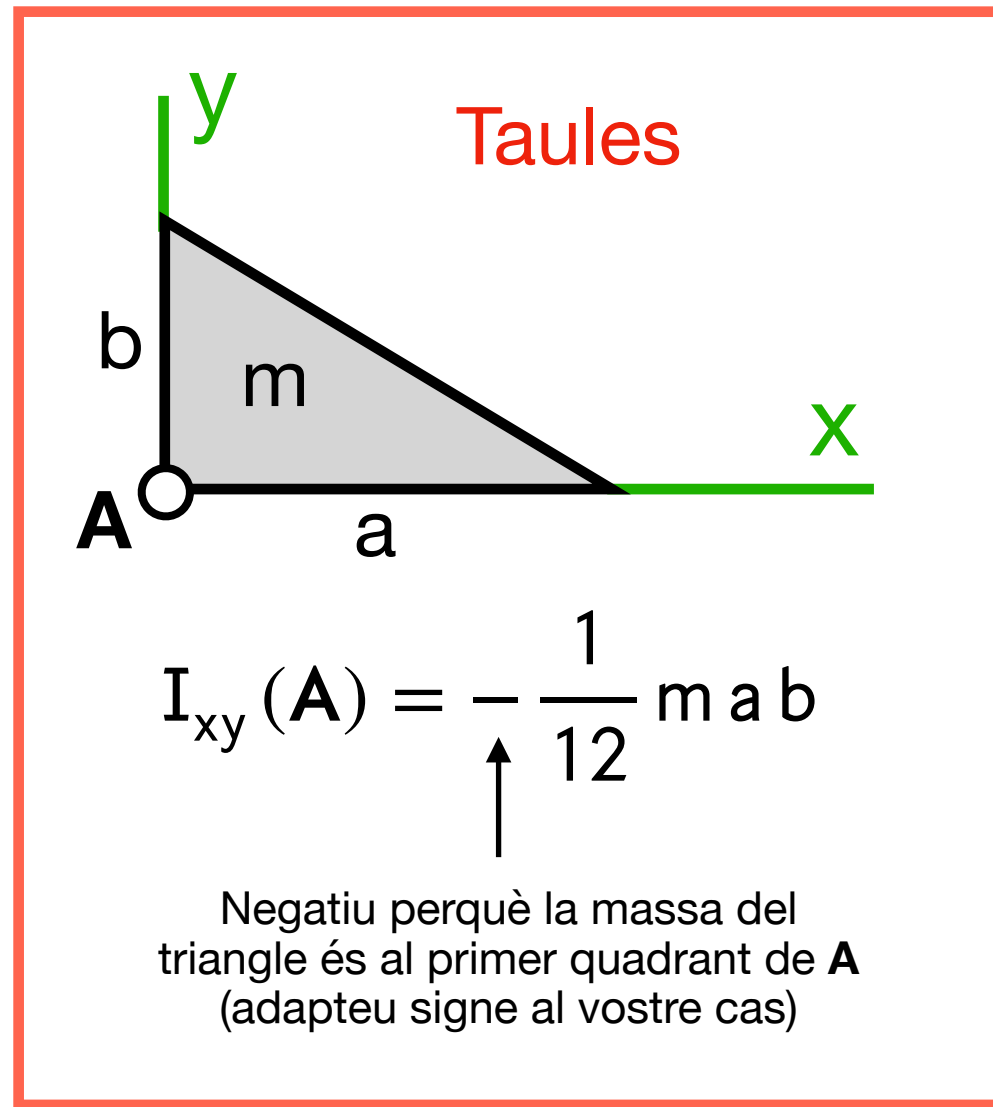
- Us caldrà I_{13} (vegeu-ne el càlcul a la següent transparència)

Solució

$$N = \frac{2}{3}mg + \frac{ac}{b}v_0 \cos \theta - \frac{3}{4}ma\dot{\theta}^2$$



$I_{13}(\mathbf{O})$?



$I_{13}(\mathbf{A})$ de taules + **doble Steiner** per passar de $\mathbf{A}$ a $\mathbf{O}$:

$$I_{13}(\mathbf{O}) = I_{13}(\mathbf{A}) + I_{13}^{\oplus}(\mathbf{O}) - I_{13}^{\oplus}(\mathbf{A})$$

$$I_{13}(\mathbf{O}) = -\frac{1}{12} m (3a) (3b) + m 2b a - (-m a b) = \frac{9}{4} m a b$$

Negatiu perquè tota la massa de la placa és al 3er quadrant de $\mathbf{A}$

Positiu perquè la massa concentrada a $\mathbf{G}$ és al 4rt quadrant de $\mathbf{O}$

Negatiu perquè la massa concentrada a $\mathbf{G}$ és al 3er quadrant de $\mathbf{A}$

Surt positiu perquè tota la massa de la placa és al 4rt quadrant de $\mathbf{O}$

DEURES

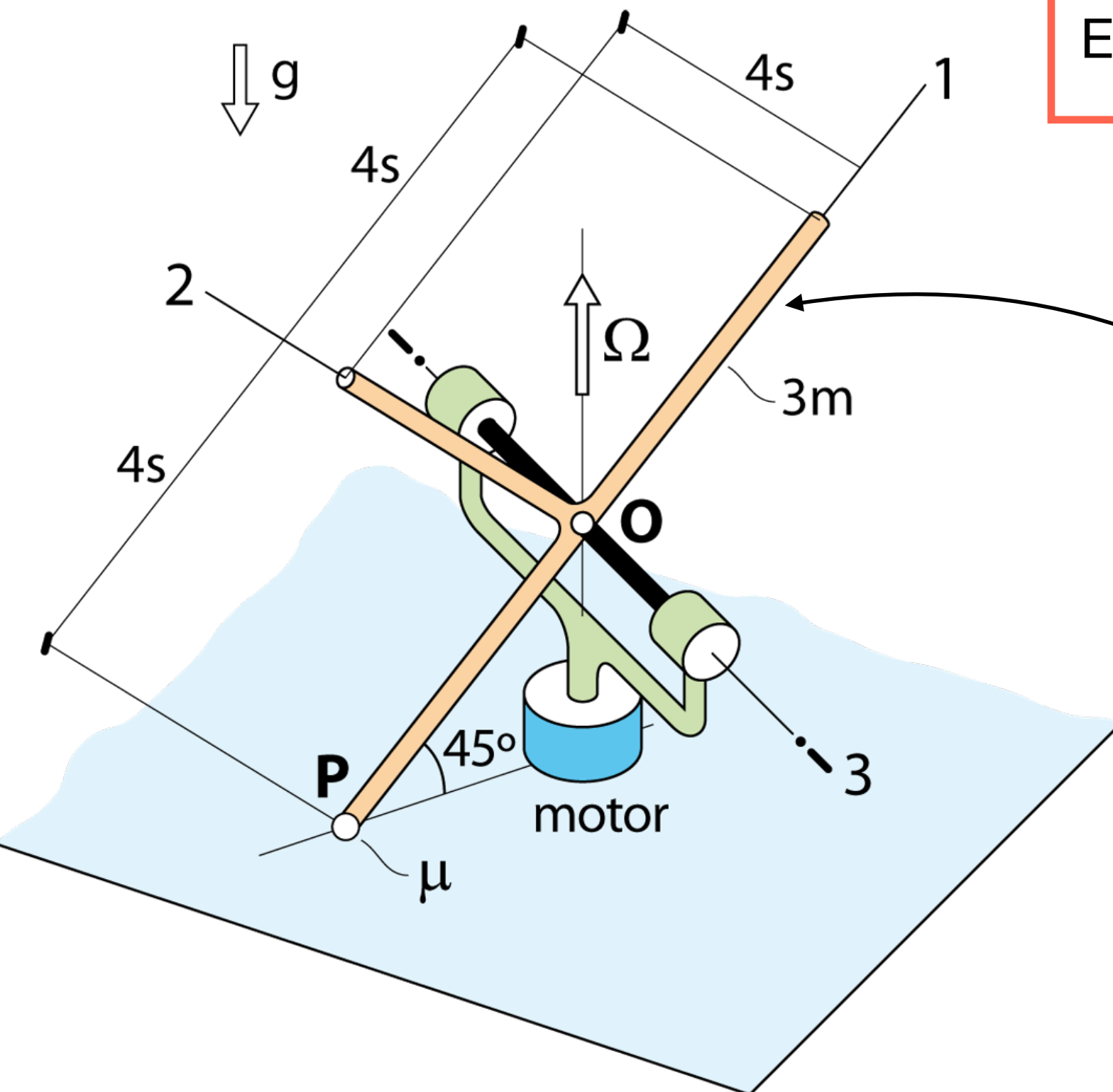
Finalment, per practicar, calculeu **TOT** el tensor d'inèrcia de la placa referit al punt **O**

$$\Omega_{\text{forq}} = \Omega = \omega$$

Forq. de massa ≈ 0

Ω_{critica} per pèrdua contacte a P?

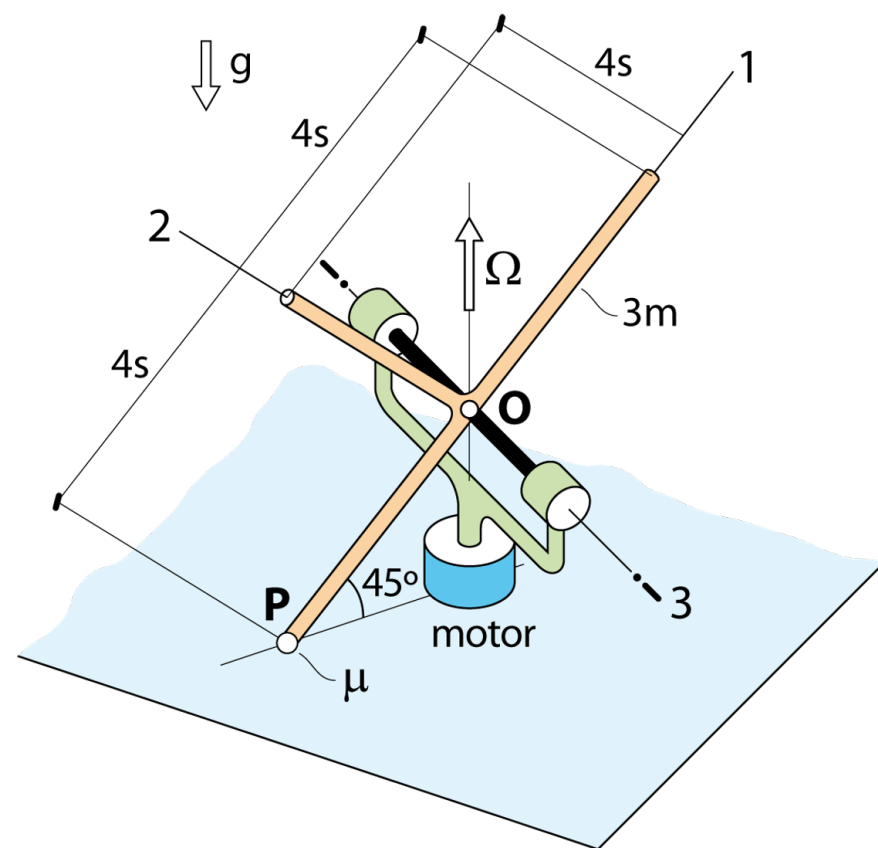
Eq. mov. quan $\Omega > \Omega_{\text{critica}}$



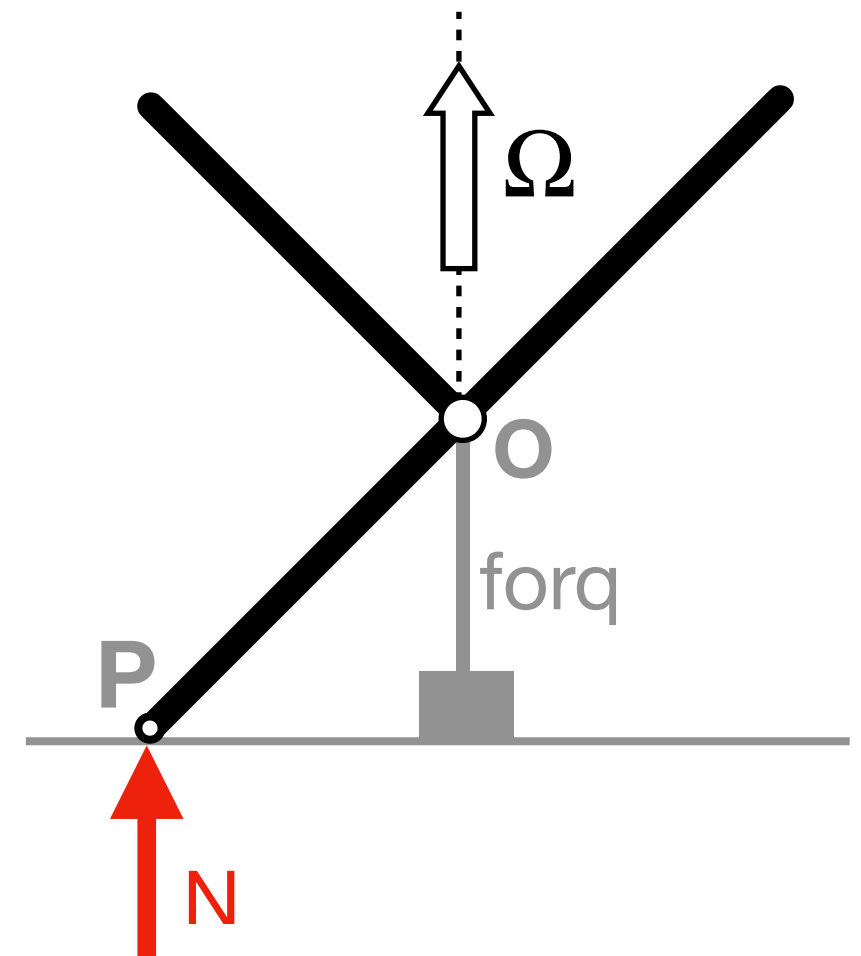
"Peça"

GL?

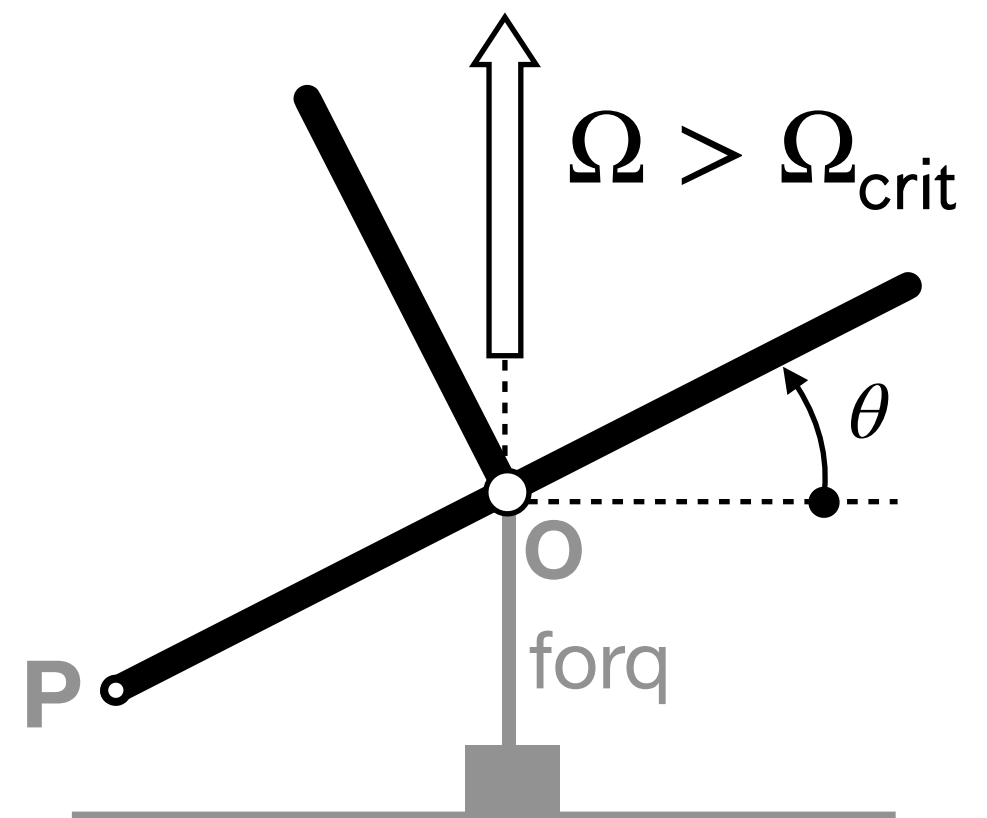
2 situacions!

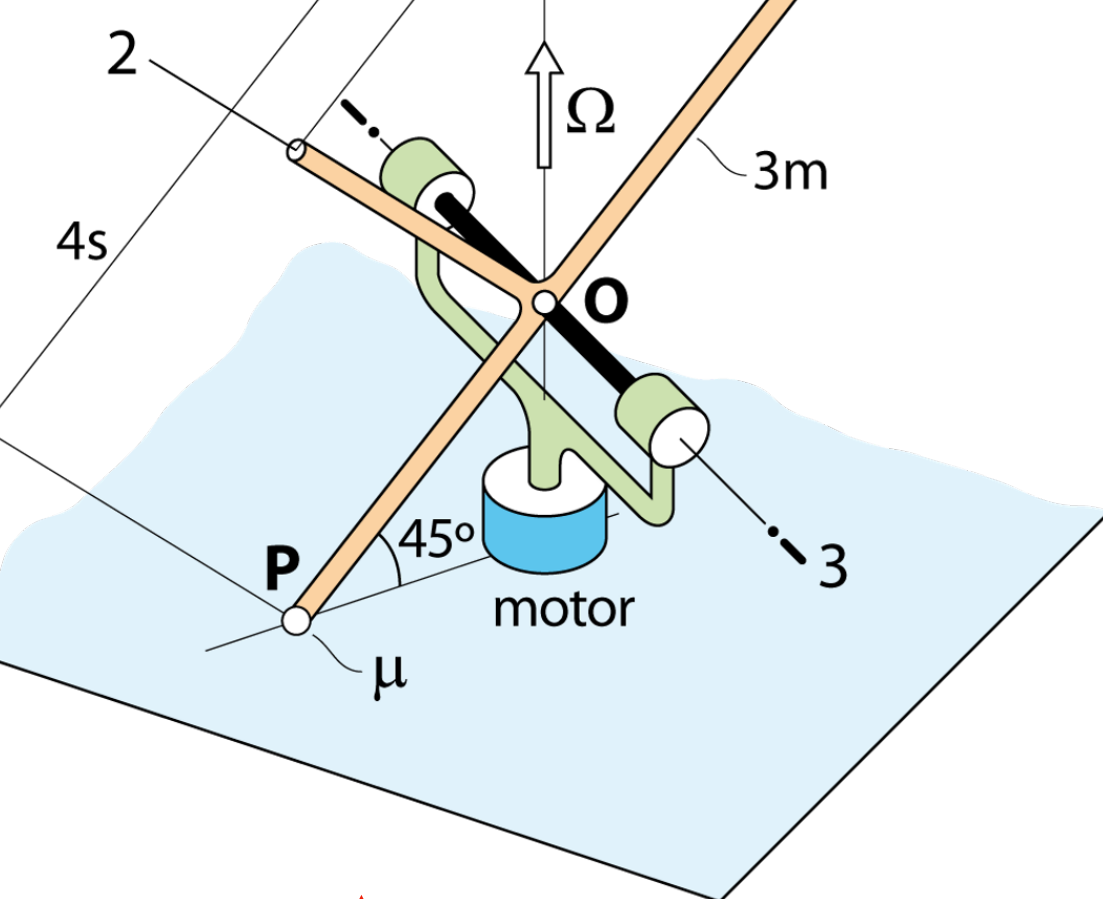


P manté
contacte
amb T



Contacte
perdut

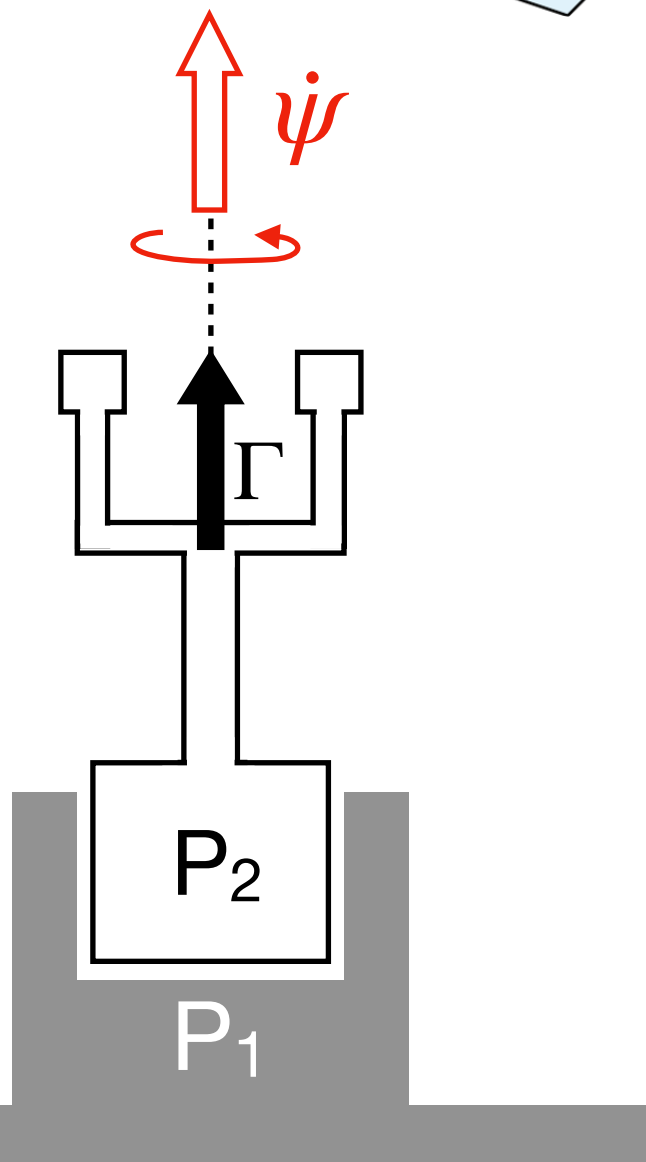




Recordeu

Γ conegut $\Rightarrow \ddot{\psi}$ és incògnita

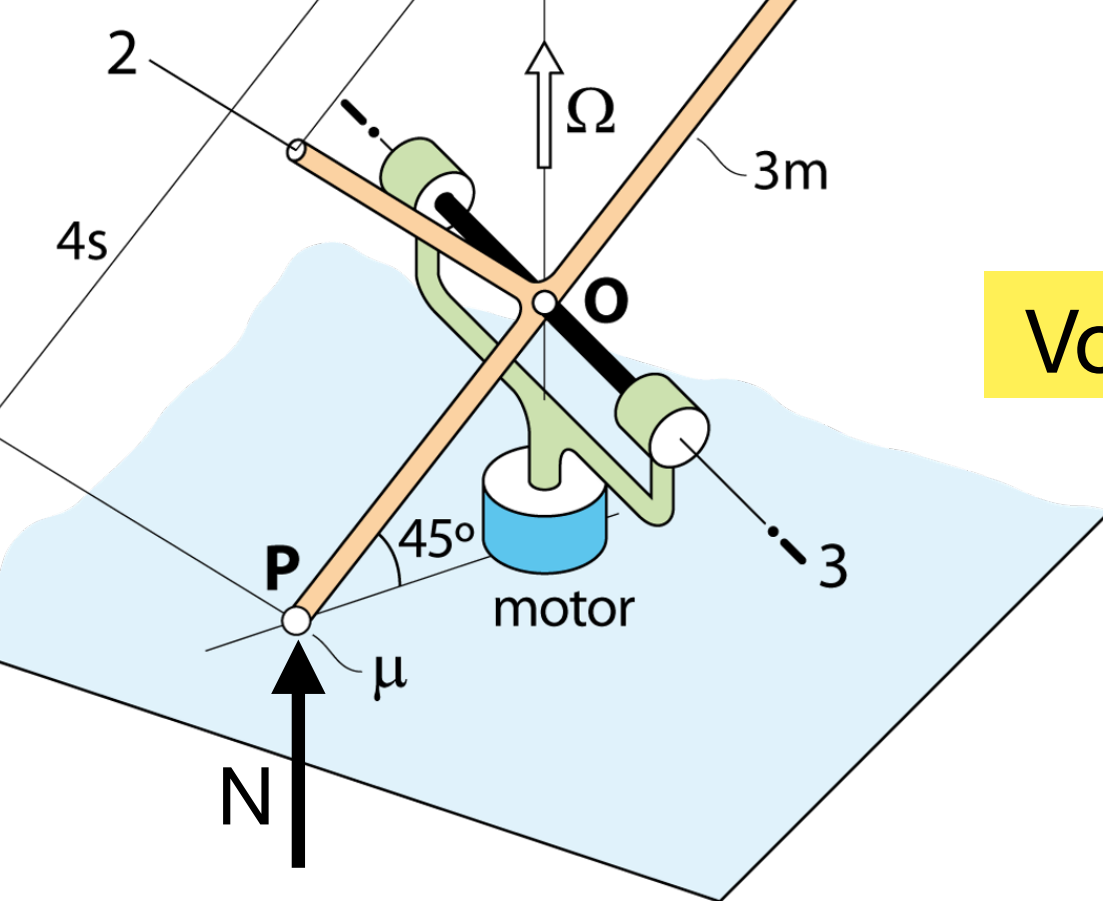
$\ddot{\psi}$ coneguda $\Rightarrow \Gamma$ és incògnita



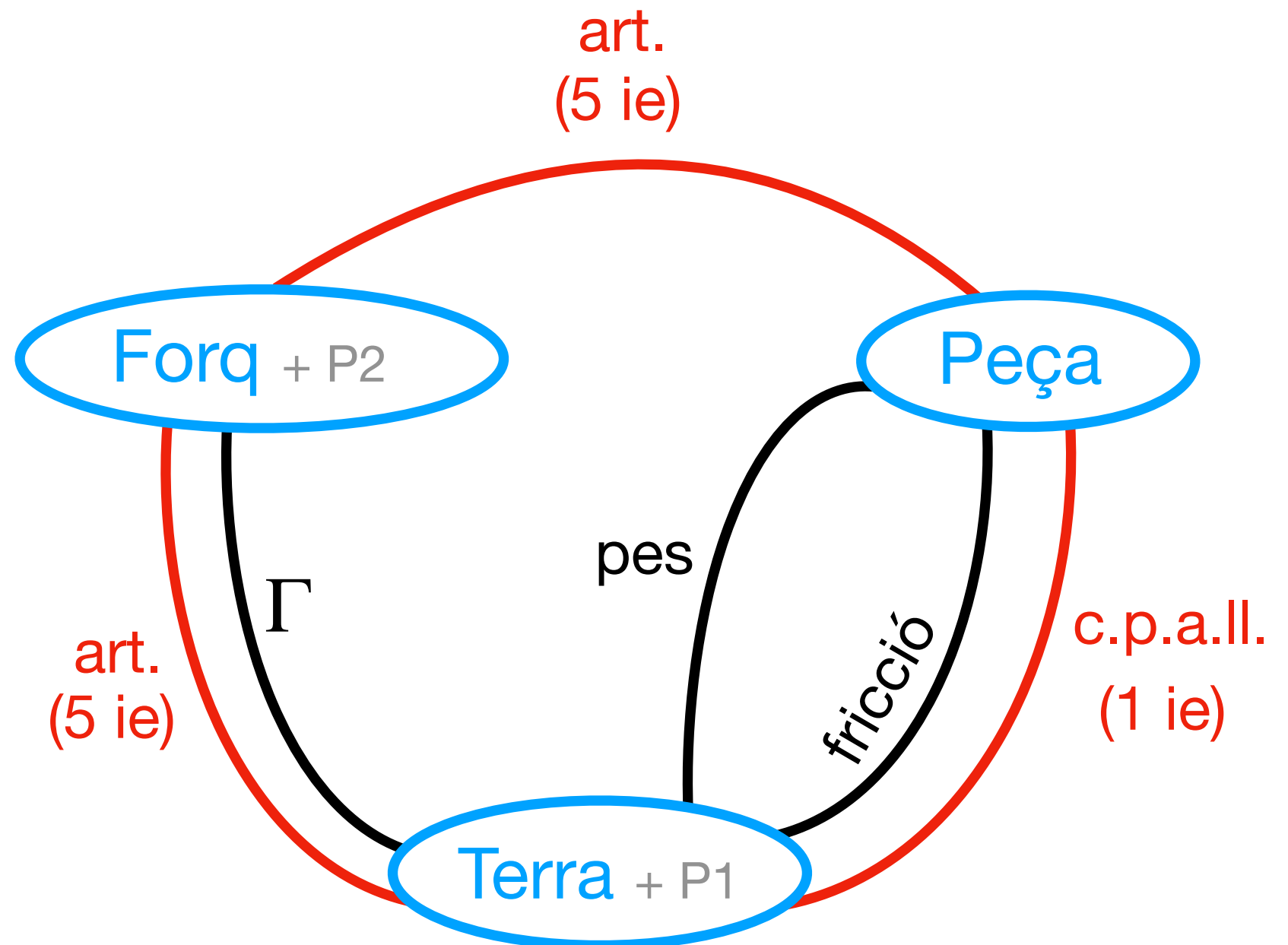
En aquest exercici

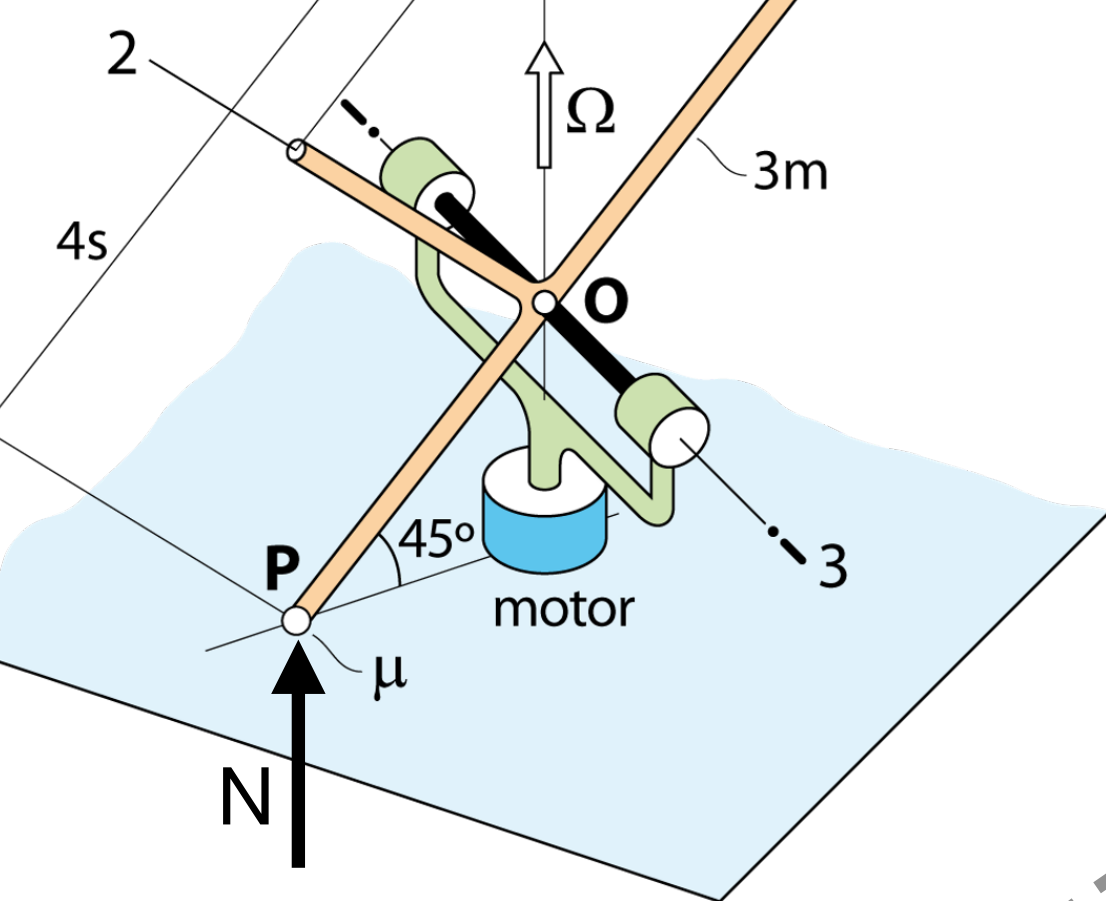
$\ddot{\psi} = \Omega = ct \Rightarrow \ddot{\psi} = 0$ (coneguda)

Γ serà incògnita

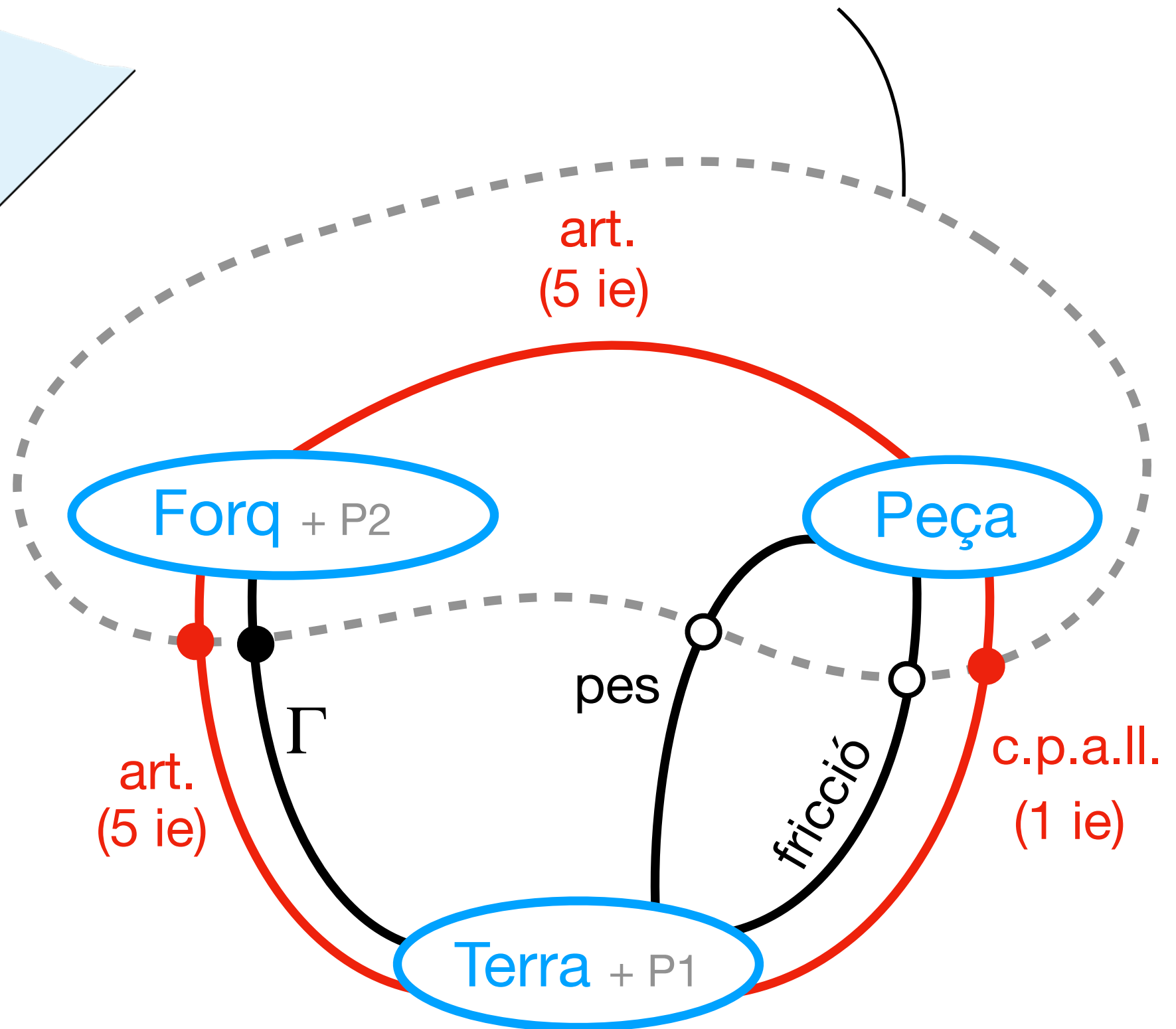


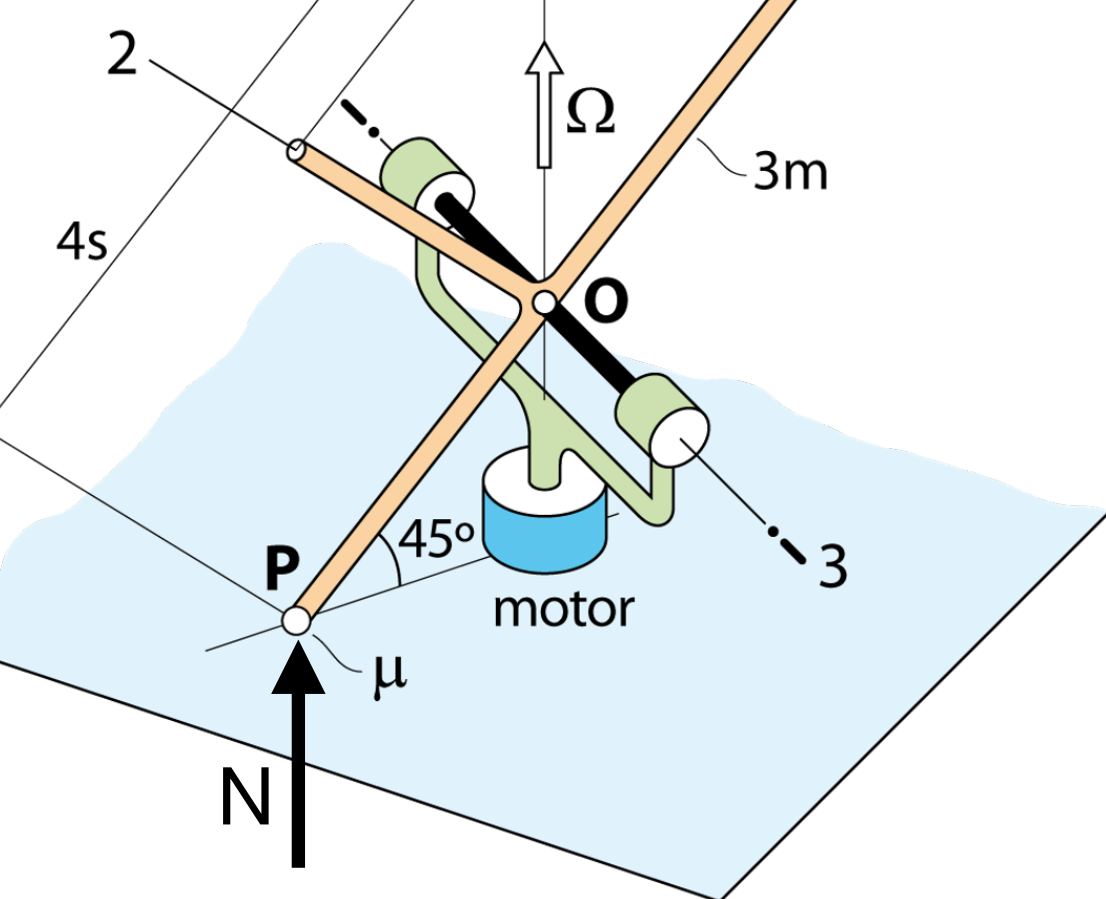
Volem $N \Rightarrow$ SIST ha d'incloure la peça!





$6 \text{ ie} + \Gamma \Rightarrow \text{INDETERMINAT}$

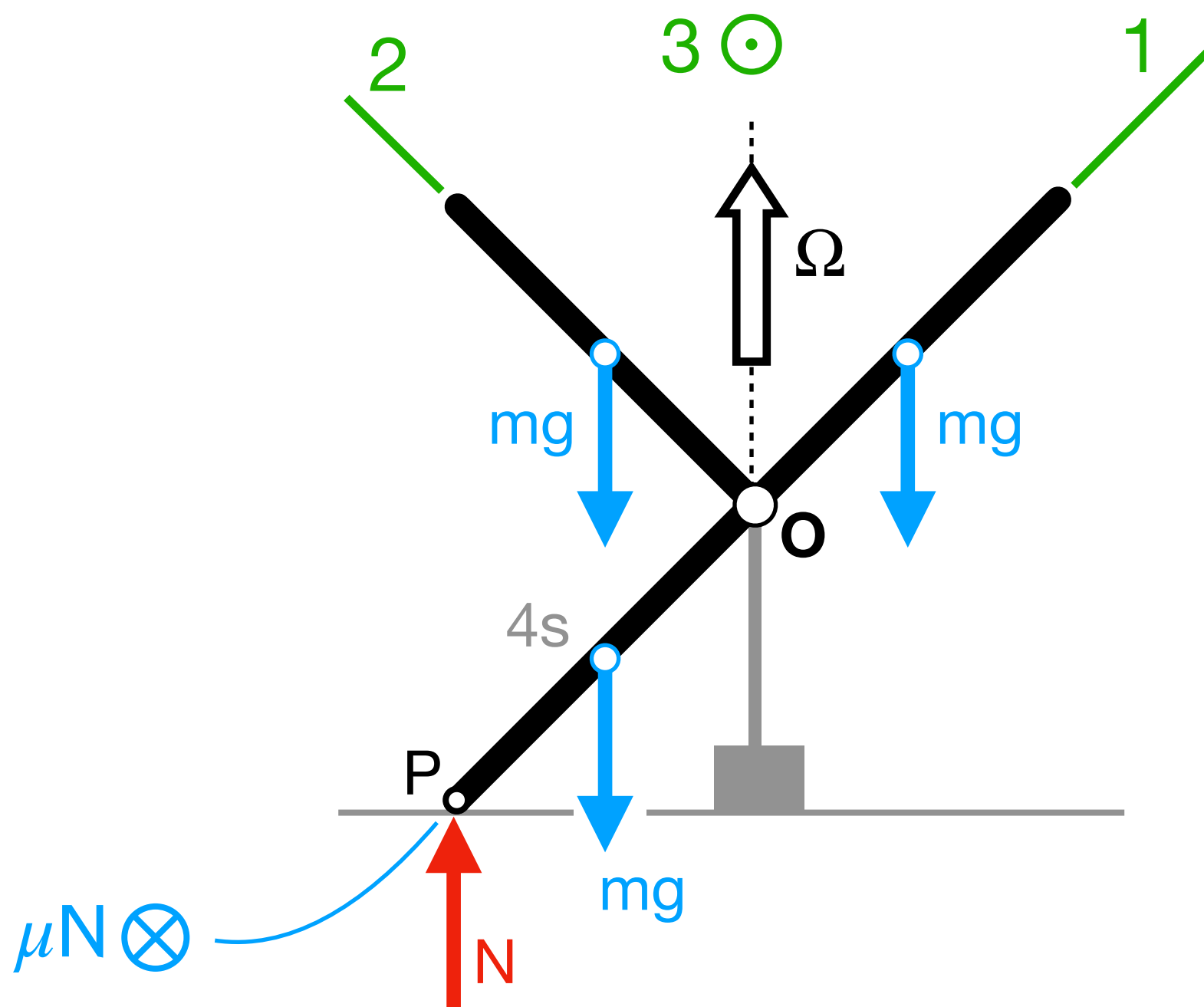




Forces sobre "Peça"

$$\left\{ \bar{\mathbf{F}}_{\text{Forq} \rightarrow \text{Peça}} \right\}_B = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

$$\left\{ \bar{\mathbf{M}}_{\text{Forq} \rightarrow \text{Peça}} (\mathbf{O}) \right\}_B = \begin{Bmatrix} M_1 \\ M_2 \\ 0 \end{Bmatrix}$$



DEURES

Determineu

- Parell motor Γ per mantenir $\Omega = ct$
- Eq. mov. per al cas en que el contacte a P ja s'ha perdut ($\Omega > \Omega_{\text{critica}}$)